



SOLUTIONS OF CURRENT ELECTRICITY

EXERCISE-1

PART - I

Section (A) :

A-1. $i = i_0 + \alpha \sin \pi t$

$$\frac{dq}{dt} = 10 + \frac{\pi}{2} \sin \pi t$$

$$q = \int_0^3 \left(10 + \frac{\pi}{2} \sin \pi t \right) dt = 10 \times 3 + \frac{\pi}{2} \times \frac{1}{\pi} \times 2 = 31 \text{ C} \quad \text{Ans.}$$

$$\text{Average current} = \frac{q}{\Delta t} = \frac{31}{3} \text{ A} \quad \text{Ans.}$$

A-2 Mass of copper per unit volume = $9 \times 10^3 \text{ kg}$

$$\text{Atoms of Cu per unit volume} = \frac{9 \times 6.023 \times 10^{29}}{63.5} = n.$$

$$i = neAV_d$$

$$V_d = \frac{1.5 \times 63.5}{1.6 \times 10^{-19} \times 10^{-7} \times 9 \times 6.023 \times 10^{29}} = 1.10 \times 10^{-3} \text{ ms}^{-1}.$$

A-3. (i) $Q = i \Delta t = 5 \times 4 \times 60 = 1200 \text{ C}$

(ii) $Q = ne \Rightarrow n = \frac{Q}{e} = \frac{1200}{1.6 \times 10^{-19}} = 75 \times 10^{20}$

Section (B)

B-1. (a) $\frac{20 \times 10^{-3}}{e} \rightarrow$ no of electrons passing per second

$$= \frac{20 \times 10^{-3}}{1.6 \times 10^{-19}} = \frac{2 \times 10^{17}}{1.6} = 1.25 \times 10^{17}.$$

(b) $j = \frac{20 \times 10^{-3}}{\pi (0.2 \times 10^{-3})^2} = \frac{1}{2\pi} \times 10^6 \text{ A/m}^2.$

B-2. Voltage the across the wire = $E\ell = 25 \times 2 = 50 \text{ V}$

$$i = \frac{50}{5} = 10 \text{ A.}$$

B-3. (i) $R_{15} = \frac{200}{10} = 20 \Omega, R_t = \frac{200}{9} \Omega$

$$R_t = R_{15} (1 + \alpha \Delta t) \Rightarrow \frac{200}{9} = 20 \left(1 + \frac{\Delta t}{234} \right)$$

$$\frac{\Delta t}{234} = \frac{1}{9} \Rightarrow \Delta t = \frac{234}{9} = 26 \Rightarrow t = 26 + 15 = 41^\circ \text{ C.}$$

(ii) $\alpha = \frac{1}{R} \frac{dR}{dt} \Rightarrow \alpha \int_0^{273} dt = \int_{10}^{20} \frac{dR}{R}$

$$\alpha \cdot 273 = \ln \left(\frac{20}{10} \right) \quad \alpha = \frac{\ln 2}{273} ^\circ \text{C}^{-1}.$$



B-4. $\frac{I}{V} = \text{slope of given graph} = \frac{1}{R}$ or $R = \frac{1}{\text{slope}}$

Resistance of a metallic wire increases with increase in temperature.

$$(\text{slope})_{T_2} < (\text{slope})_{T_1}$$

$$\therefore \frac{1}{(\text{slope})_{T_2}} > \frac{1}{(\text{slope})_{T_1}}$$

or $R_{T_2} > R_{T_1}$

or $T_2 > T_1$

B-5. Resistance $R = \frac{\rho \ell}{A}$

By partial differentiation

$$\Rightarrow \frac{\Delta R}{R} = \frac{\Delta \ell}{\ell} - \frac{\Delta A}{A} \quad \dots\dots\dots(1)$$

$$\therefore \rho = \text{Constant}$$

$\therefore$ Volume of wire remains constant

$$A \times \ell = \text{Constant}$$

By partial differentiation

$$\Rightarrow \frac{\Delta A}{A} + \frac{\Delta \ell}{\ell} = 0 \quad \dots\dots\dots(2)$$

By equation (1) and (2)

We get $\frac{\Delta R}{R} = 2 \frac{\Delta \ell}{\ell}$

$$\Rightarrow (\% \text{ change in } R) = 2 (\% \text{ change in length}) = 0.2 \%$$

Ans.

B-6 (i) $R = \frac{\rho \ell}{A} = \frac{(3.5 \times 10^{-5}) \times (50 \times 10^{-2})}{(1.0 \times 10^{-2})^2} = \frac{0.35}{2} = 0.175 \Omega$

(ii) $R = \frac{\rho \ell}{A} = \frac{(3.5 \times 10^{-5}) (1.0 \times 10^{-2})}{(1.0 \times 10^{-2}) (50 \times 10^{-2})} = 7 \times 10^{-5} \Omega$

Section (C)

C-1. (a) In each case E.M.F. = 10 V

(b) For case (A), battery is providing current to the circuit hence acting as source.

For case (B) battery is taking current from external source, hence acting as a load.

(c) For case (A) $V_A = E - ir = 10 - 1 \times 1 = 9 \text{ V}$

For case (B) $V_B = E + ir = 10 + 1 \times 1 = 11 \text{ V}$

(d) For case (A) $P_A = V_A i = 9 \times 1 = 9 \text{ W}$

For case (B) $P_B = V_B i = 11 \times 1 = 11 \text{ W}$

(e) For case (A) $\frac{H_A}{t} = i_A^2 r_A = (1)^2 \times 1 = 1 \text{ W}$

For case (B) $\frac{H_B}{t} = i_B^2 r_B = (1)^2 \times 1 = 1 \text{ W}$

(f) For case (A) $P_A = E_A i = 10 \times 1 = 10 \text{ W}$

For case (B) $P_B = E_B i = 10 \times 1 = 10 \text{ W}$

(g) For case (A) $V_{\text{Box}} = V_A = 9 \text{ V}$

For case (B) $V_{\text{Box}} = V_B = 11 \text{ V}$

(h) For case (A) $P_{\text{Box}} = -9 \times 1 = -9 \text{ W}$

For case (B) $P_{\text{Box}} = 11 \times 1 = 11 \text{ W}$



C-2. $V = \frac{W}{O}$

$$Q = I \times t$$
$$= 3 \times 12 = 36 \text{ C}$$

$$V = \frac{500}{36} = \frac{125}{9} \text{ V Ans.}$$

- C-3.** (a) equal in all point a, b, and c
 (b) $V_b > V_c = V_a$
 (c) $(P.E)_a = (P.E)_c > (P.E)_b$
 $\therefore P.E = -eV$

C-4. (a) $V = E - ir = 12 - 90 \times 5 \times 10^{-2} = 12 - 4.5 = 7.5 \text{ V}$

$$(b) I_{\max} = \frac{E}{r'} = \frac{12}{500} = 24 \text{ mA}$$

(c) For charging of battery $V = E + ir$, $V > E \Rightarrow V > 12 \text{ V}$

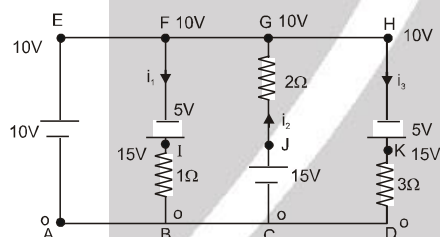
C-5. (a) $i = \frac{P}{V} = \frac{1000}{220} = \frac{50}{11} = 4.55 \text{ A}$

$$(b) R = \frac{V^2}{P} = \frac{(220)^2}{1000} = \frac{22 \times 11}{5} = 48.4 \Omega$$

(c) $P = 1 \text{ kW}$

$$(d) H = \frac{Q}{t} = \frac{1000}{4.2} = 240 \text{ cal/sec}$$

$$(e) \text{ tH} = \text{mL} \quad \Rightarrow \quad m = \frac{\text{Ht}}{l} = \frac{240 \times 60}{540} = \frac{80}{3} \text{ gm.}$$



- (a) $V_A = V_B = V_C = V_D = 0$ $V_E = V_F = V_G = V_H = 10V$
 $V_I = 10 + 5 = 15V$ $V_J = 15V$ $V_K = 10 + 5 = 15V$
- (b) $V_{BI} = 15V$, $V_{JG} = 5V$, $V_{KD} = 15V$
- (c) Each battery is supplying the current hence each battery is acting as a source.
- (d) Let current through BF, CG, HP is respectively i_1, i_2, i_3

$$i_1 = \frac{15}{1} = 15 \text{ amp } \downarrow, \quad i_2 = \frac{5}{2} = 2.5 \text{ amp } \uparrow \quad i_3 = \frac{15}{3} = 5 \text{ amp } \downarrow$$

For 10 V Battery , current = $i_1 - i_2 + i_3 = 15 - 2.5 + 5 = 17.5 \text{ amp} \uparrow$

$$(e) P_{1\Omega} = \frac{V^2}{R} = \frac{(15)^2}{1} = 225 \text{ W} \quad P_{2\Omega} = \frac{(5)^2}{2} = 12.5 \text{ W} \quad P_{3\Omega} = \frac{(15)^2}{3} = \frac{225}{3} = 75 \text{ W}$$

Hence, 1Ω resistance consumes the maximum power.

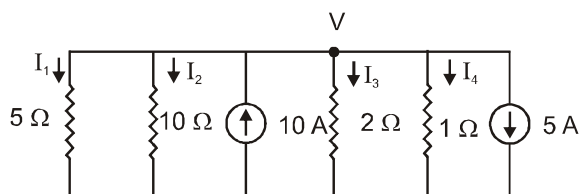
(f) $P_I = E_I i_1 = 10 \times 17.5 = 175 \text{ W}$ $P_{II} = E_2 i_1 = 5 \times 15 = 75 \text{ W}$

$$P_{III} = E_3 i_2 = 15 \times 2.5 = 37.5 \text{ W} \qquad P_{IV} = E_4 i_3 = 5 \times 5 = 25 \text{ W}$$

hence left most battery consume maximum power.



- C-7.** The circuit shown above is a parallel circuit, and consists of a single node. By assuming voltage V at the node, we can find out the current in $10\ \Omega$ branch.



According to Kirchhoff's Current Law,

$$I_1 + I_2 + I_3 + I_4 + 5 = 10$$

By using Ohm's law, we have

$$I_1 = \frac{V}{5}; I_2 = \frac{V}{10}; I_3 = \frac{V}{2}; I_4 = \frac{V}{1}$$

$$\frac{V}{5} + \frac{V}{10} + \frac{V}{2} + V + 5 = 10$$

$$V \left[\frac{1}{5} + \frac{1}{10} + \frac{1}{2} + 1 \right] = 5$$

$$V [0.2 + 0.1 + 0.5 + 1] = 5$$

$$V = \frac{5}{1.8} = 2.78\text{ V Ans}$$

$\therefore$ The voltage across $10\ \Omega$ resistor is 2.78 V and the current passing through it is

$$I_2 = \frac{V}{10} = \frac{2.78}{10} = 0.278\text{ A Ans}$$

- C-8.** According to Kirchhoff's Voltage Law, the sum of the potential drops equal to the sum of the potential rises;

Therefore, $30 = 2 + 1 + V_1 + 3 + 5$

or $V_1 = 30 - 11 = 19\text{ V Ans.}$

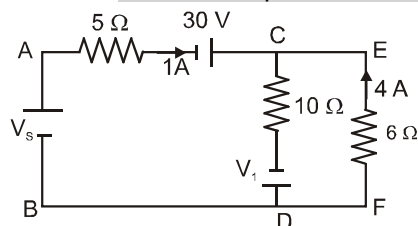
- C-9.** (a) $H = I^2 R \Delta t \Rightarrow 400 = 2^2 \times R \times 10 \Rightarrow R = 10\ \Omega$
 (b) $H = I^2 R \Delta t$ Now $I = 4\text{ A}$, $\Delta t = 20\text{ s} \Rightarrow H = 4^2 \times 10 \times 20 = 3200\text{ J}$

- C-10.** By applying Kirchhoff's Current Law, we get the current in $10\ \Omega$

$$I_{10} = I_5 + I_6 = 4 + 1 = 5\text{ A Ans.}$$

The voltage across $6\ \Omega$ resistor is $V_6 = 24\text{ V} = V_1 - 10 \times 5 \Rightarrow V_1 = 74\text{ V}$

Now, consider the loop ABFE



If we apply Kirchhoff's Voltage Law, we get

$$V_s = 5 - 30 - 24 = -49\text{ V}$$



Section : (D)

D-1. $R = \frac{(220)^2}{500}$ for the two bulbs,

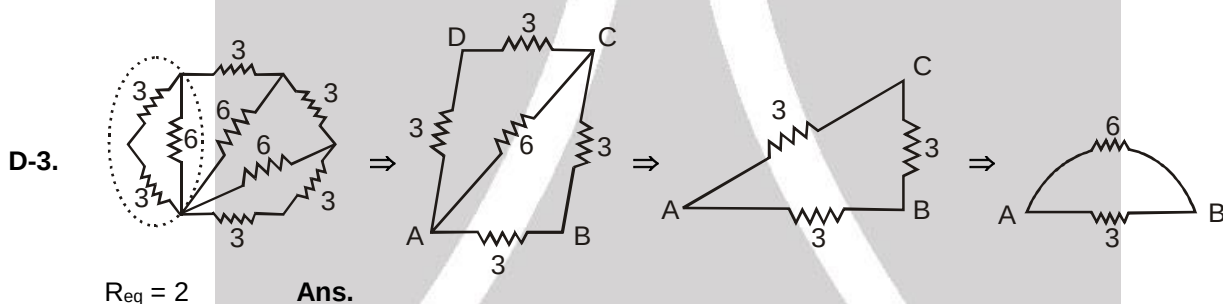
$\frac{110}{2}$ V will be the potential difference individually

$P = \frac{1}{4} \times \frac{(110)^2 \times (500)}{(220)^2} = \frac{500}{16} = \frac{125}{4} = 31.25 \text{ W}$ **Ans.**

- D-2.** For quantitative purposes we assume that the resistances of the bulbs do not depend upon the voltages across them. This is far from accurate, but will give the correct qualitative conclusion. If the (r.m.s.) supply voltage is V , the resistance r_i of a bulb is V^2/w_i , where w_i is the nominal rating of the bulb. When the two bulbs are connected in series across the supply, the (r.m.s) current drawn is $V/(r_A + r_B)$ and the power dissipated in bulb i ($i = A$ or B) is

$$P_i = \frac{V^2}{w_i} \left[\frac{V}{(V^2/w_A) + (V^2/w_B)} \right]^2$$

According to the original agreement ($w_A = w_B = 100 \text{ W}$), both P_A and P_B should be 25 W . Actually, $P_A = 8 \text{ W}$ and $P_B = 32 \text{ W}$, and so A clearly failed his examinations. By comparison, student B might be considered a double winner : he gets 32 W , but pays for only $(8 + 32)/2 = 20 \text{ W}$. On the other hand, 32 W is still a very poor light to study by and B also could well have failed his examinations.



D-4. (a) $R_{eq} = R_1 + R_2 + R_3 + r = 10 \Omega$

(b) $i = \frac{\varepsilon}{R_{eq}} = \frac{10}{10} = 1 \text{ A}$

(c) $V_{3\Omega} = 1 \times 3 = 3 \text{ V}$, $V_{4\Omega} = 1 \times 4 = 4 \text{ V}$, $V_{2\Omega} = 1 \times 2 = 2 \text{ V}$

(d) $P_{consumed} = \varepsilon i = 10 \times 1 = 10 \text{ W}$

(e) $P_{generated} = i^2 r = 1 \text{ W}$ inside the battery

(f) $P_{output} = (\varepsilon - ir)i = 10 - 1 = 9 \text{ W}$

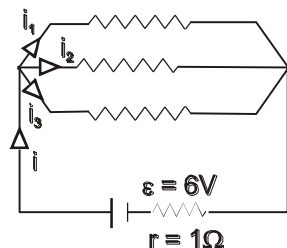
(g) $V_{battery} = \varepsilon - ir = 9 \text{ V}$

(h) In series, $P \propto R$ 4Ω consumes maximum power

(i) $P_{3\Omega} = i^2 R = 3 \text{ W}$.



- D-5.** (a) $R_{eq} = 2 + 1 = 3 \Omega$
 (b) $i = \frac{\varepsilon}{R_{eq}} = \frac{6}{3} = 2 \text{ A}$



$$i_1 + i_2 + i_3 = 2 \quad i_1 : i_2 : i_3 = \frac{1}{8} : \frac{1}{4} : \frac{1}{8} = 1 : 2 : 1 \quad i_1 = i_3 = \frac{1}{2} \text{ A}, \quad i_2 = 1 \text{ A}$$

- (c) $V_{\text{across battery}} = \varepsilon - ir = 6 - 2 \times 1 = 4 \text{ V}$ $V_{\text{across each cell}} = 4 \text{ V}$
 (d) P of the cell consumed $P = \varepsilon i = 12 \text{ W}$

- (e) P heat generated in cell $P = i^2 r = 4 \text{ W}$

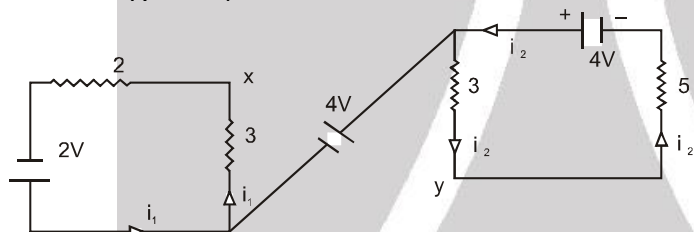
- (f) $P_{\text{output}} = \varepsilon i - i^2 r = 8 \text{ W}$

- (g) In parallel, $P \propto \frac{1}{R}$

4Ω consumes max power

- (h) $P_{4\Omega} = \frac{V^2}{R} = \frac{4 \times 4}{4} = 4 \text{ W}$

D-6.



$$i_1 = \frac{2}{5} = 0.4 \text{ A} \quad i_2 = \frac{4}{8} = 0.5 \text{ A}$$

- (a) $V_x + 3i_1 + 4 - 3i_2 = V_y$

$$V_x - V_y = -4 + 3 \times 0.5 - 3 \times 0.4 = -4 + 1.5 - 1.2 = -3.7 \text{ V} \Rightarrow \text{potential difference} = 3.7 \text{ V.}$$

- (b) Still same as No current flows in that cell

D-7.

(i) $R_{AB} = \frac{5 \times 1}{5 + 1} = \frac{5}{6} \Omega$

(ii) $R_{CD} = \frac{\left(\frac{3 \times 6}{3 + 6}\right) \times 6}{\left(\frac{3 \times 6}{3 + 6}\right) + 6} = \frac{2 \times 6}{2 + 6} = \frac{3}{2} = 1.5 \Omega$

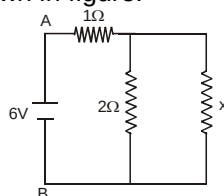
(iii) $R_{EF} = \frac{\left[\left(\frac{3 \times 6}{3 + 6}\right) + 4\right] \times 2}{\left(\frac{3 \times 6}{3 + 6}\right) + 4 + 2} = \frac{(2 + 4) \cdot 2}{2 + 4 + 2} = \frac{3}{2} = 1.5 \Omega$

(iv) $R_{AF} = R_{AB} = \frac{5}{6} \Omega$

(v) $R_{AC} = \frac{\left(\frac{6 \times 6}{6 + 6} + 1\right) \cdot 2}{\left(\frac{6 \times 6}{6 + 6} + 1\right) + 2} = \frac{(3 + 1) \cdot 2}{(3 + 1) + 2} = \frac{4}{3} \Omega$



- D-8.** (i) Let $R_{AB} = x$. Then, we can break one chain and connect a resistance of magnitude x in place of it. Thus, the circuit remains as shown in figure.



Now, 2Ω and x are in parallel. So, their combined resistance is $\frac{2x}{2+x}$

or $R_{AB} = 1 + \frac{2x}{2+x}$

But R_{AB} is assumed to be x . Therefore,

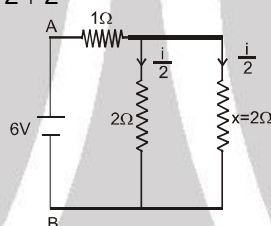
$$x = 1 + \frac{2x}{2+x}$$

Solving this equation, we get

$$x = 2\Omega$$

Hence proved.

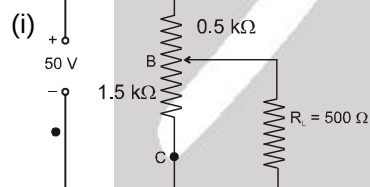
- (ii) Net resistance of circuit $R = 1 + \frac{2 \times 2}{2+2} = 2\Omega$



$\therefore$ Current through battery $i = \frac{6}{2} = 3A$

This current is equally distributed in 2Ω and 2Ω resistances. Therefore, the desired current is $\frac{i}{2}$ or $1.5A$.

D-9.



As $V \propto R$ $V_2 = V_{BC} = \left(\frac{R_{BC}}{R_{AB} + R_{BC}} \right) V = \left(\frac{1.5 \times 0.5}{0.5 + \frac{1.5 \times 0.5}{1.5 + 0.5}} \right) 50$

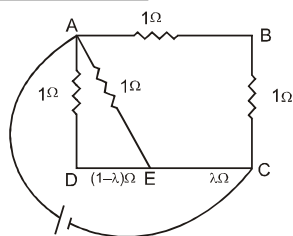
$$= \left(\frac{0.75}{1 + 0.75} \right) 50 = \frac{150}{7} = 21.43 V$$

(ii) $V_{BC} = 40 V \Rightarrow V_{BC} = \left(\frac{R_{BC}}{R_{BC} + R_{AC}} \right) V$

$$\Rightarrow 40 = \frac{R_{BC}}{2000} 50 \Rightarrow R_{BC} = 1600 \Omega$$



D-10.



$$V_B = V_E \Rightarrow \frac{1 \times (2 - \lambda)}{3 - \lambda} = \lambda$$

$$2 - \lambda = 3\lambda - \lambda^2 \quad \lambda^2 - 4\lambda + 2 = 0 \quad \lambda = \frac{4 \pm \sqrt{16 - 8}}{2}$$

$$\lambda = 2 - \sqrt{2} \quad (\because \lambda < 2)$$

$$\frac{CE}{DE} = \frac{R_{CE}}{R_{DE}} = \frac{\lambda}{1 - \lambda} = \frac{2 - \sqrt{2}}{1 - 2 + \sqrt{2}} = \frac{2 - \sqrt{2}}{\sqrt{2} - 1} = \frac{\sqrt{2}(\sqrt{2} - 1)}{\sqrt{2} - 1} = \sqrt{2} : 1$$

D-11. For maximum resistance, all resistance are in series

$$R_{\max} = 20 + 50 + 100 = 170 \, \Omega$$

For minimum resistance, all resistance are in parallel

$$\frac{1}{R_{\min}} = \frac{1}{100} + \frac{1}{50} + \frac{1}{20} = \frac{1 + 2 + 5}{100} = \frac{8}{100}$$

$$R_{\min} = \frac{100}{8} = 12.5 \, \Omega$$

$$\text{D-12. case (a) } R_{\text{eq}} = \frac{180}{3} = 60 \, \Omega \quad i = \frac{60}{60} = 1 \text{ amp}$$

$$\text{case (b) } R_{\text{eq}} = \frac{180}{2} = 90 \, \Omega \quad i = \frac{60}{90} = \frac{2}{3} \text{ amp}$$

$$\text{case (c) } R_{\text{eq}} = 180 \, \Omega \quad i = \frac{60}{180} = \frac{1}{3} \text{ amp}$$

D-13. (a) S is open

$$i = \frac{3}{30} = 0.1 \text{ A} \quad \text{Ans.}$$

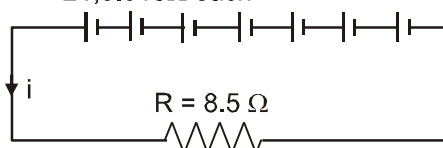
(b) S is close

$$i = \frac{3}{10} = 0.3 \text{ A} \quad \text{Ans.}$$

Section (E)

2V, 0.015Ω each

E-1.



$$(i) \text{ current } i = \frac{6E}{R + 6r} = \frac{6 \times 2}{8.5 + 6 \times 0.015} = \frac{12}{8.59} = 1.4 \text{ A}$$

$$(ii) \text{ terminal voltage } V = iR = \frac{12}{8.59} \times 8.5 = 11.9 \text{ V}$$



E-2. (i) Effective emf = $4 \times 1.5 - 1.5 = 4.5 \text{ V}$

(ii) Effective internal resistance

$$r_{eq} = \frac{4 \times 0.4}{2} + 0.4 = 1.2 \Omega$$

$$R_{eq} = 6 + 1.8 + 1.2 = 9 \Omega$$

$$I = \frac{E}{R_{eq}} = \frac{4.5}{9} = \frac{1}{2} \text{ amp}$$

$$I_{36} = \frac{R_{eq}}{R_{36}} I = \frac{6}{36} \times \frac{1}{2} = \frac{1}{12}$$

$$(ii) V_{AB} = E + ir = 1.5 + \frac{1}{2} \times 0.4 = 1.5 + 0.2 = 1.7 \text{ V}$$

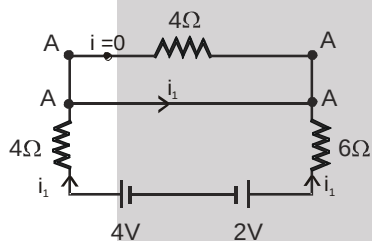
E-3. $E_{eq} = \frac{\frac{6}{600} + \frac{3}{400}}{\frac{1}{600} + \frac{1}{400}} = \frac{12+9}{2+3} = \frac{21}{5} = 4.2 \text{ V}$ **Ans**

$$\frac{1}{r_{eq}} = \frac{1}{600} + \frac{1}{400} = \frac{2+3}{1200} = \frac{5}{1200} \Rightarrow r_{eq} = 240 \Omega$$

short circuit current in AB

$$i = \frac{E_{eq}}{r_{eq}} = \frac{21/5}{240} = \frac{21}{5 \times 240} = 17.5 \times 10^{-3} \text{ amp} \Rightarrow i = 17.5 \text{ mA (from B to A)}$$

E-4.



Potential difference across upper 4Ω resistance is zero so current is also zero.

Other two resistors are in series combination so current is same

$$= \frac{4-2}{4+6} = 0.2 \text{ A.}$$

Ans.

E-5. (a) $i_1 = \frac{12}{2+0.1} = \frac{12}{2.1}$

$$i_2 = \frac{6}{0.5+0.1} = \frac{6}{0.6} = 10 \text{ A} \Rightarrow \frac{i_1}{i_2} = \frac{12}{2.1 \times 10} = 0.57$$
 Ans.

(b) $i_1 = \frac{12}{2+1} = 4 \text{ A}$

$$i_2 = \frac{6}{0.5+1} = 4 \text{ A} \Rightarrow \frac{i_1}{i_2} = 1$$
 Ans.

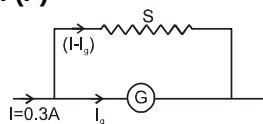
(c) $i_1 = \frac{12}{2+10} = \frac{12}{12} = 1 \text{ A}$ & $i_2 = \frac{6}{0.5+10} = \frac{6}{10.5}$

$$\Rightarrow \frac{i_1}{i_2} = \frac{1 \times 10.5}{6} = 1.75$$
 Ans.



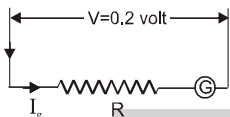
Section (F)

F-1.



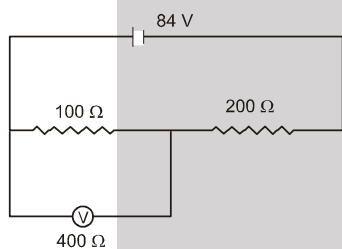
$$(I - I_g) S = I_g G \Rightarrow S = \left(\frac{I_g}{I - I_g} \right) G$$

$$S = \left(\frac{0.002}{0.3 - 0.002} \right) 30 = \frac{0.002}{0.298} \times 30 \quad S = \frac{30}{149} = 0.2013 \, \Omega$$



$$V = I_g (R + G) \Rightarrow 0.2 = 0.002 (R + 30) \quad R = 70 \, \Omega$$

F-2.

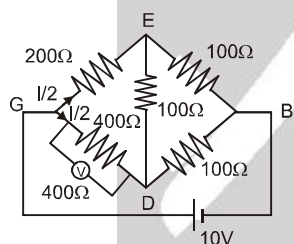


$$R_{\text{eff}} = \frac{400 \times 100}{500} + 200 \quad R_{\text{eff}} = 280 \, \Omega$$

$$i = \frac{84}{280} \, \text{A} \quad V = \frac{84}{280} \times \frac{100 \times 400}{500} = 24 \, \text{V}$$

$$(b) i = \frac{84}{300} \quad V = \frac{84}{300} \times 100 = 28 \, \text{V}$$

F-3.



$$R_{GD} = \frac{400 \times 400}{400 + 400} = 200 \, \Omega$$

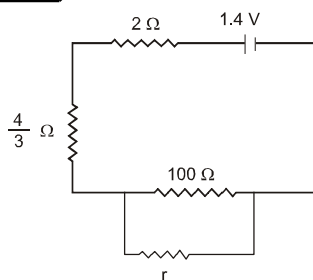
$$\text{Since } \frac{R_{GE}}{R_{GD}} = \frac{R_{EB}}{R_{DB}} \text{ it is the case of balanced Wheatstone bridge.}$$

$$R_{\text{eq}} = R_{GB} = \frac{300 \times 300}{300 + 300} = 150 \, \Omega, \quad \text{current } I = \frac{V}{R_{GB}} = \frac{10}{150} = \frac{1}{15} \, \text{amp}$$

$$\text{Potential difference across voltmeter.} = \frac{I}{30} \times 200 = \frac{1}{15} \times 100 = \frac{20}{3} \, \text{volt}$$



F-4. (i)



$$(ii) \quad R_{eq} = 2 + \frac{4}{3} + \frac{100r}{100+r} = \frac{600+6r+400+4r+300r}{3(100+r)} = \frac{310r+1000}{3(100+r)}$$

$$i = \frac{V}{R}, \quad 0.02 = \frac{1.40}{\frac{310r+1000}{3(100+r)}} \times (300+3r)$$

$$310r+1000 = 21000+210r$$

$$10r = 2000$$

$$r = 200 \Omega \quad \text{Ans.}$$

$$(iii) \quad V = ir = 0.02 \times \frac{200 \times 100}{200+100} = \frac{4}{3} = 1.33$$

$$\text{Zero error} = 1.1 - 1.33 = -0.23 \text{ V} \quad \text{Ans.}$$

F-5. Current in primary circuit. $I = \frac{\varepsilon}{9r+r} = \frac{\varepsilon}{10r}$ Potential drop across length AB = $V_{AB} = I.R$

$$V_{AB} = \frac{\varepsilon}{10r} \cdot 9r = \frac{9\varepsilon}{10}, \quad x = \frac{V_{AB}}{L} = \frac{9}{10} \frac{\varepsilon}{L}$$

$$\text{For balance point} \quad \frac{\varepsilon}{2} = x\ell = \frac{9\varepsilon}{10L} \ell, \quad \ell = \frac{5}{9} L$$

$$F-6. \quad r = \frac{R(\ell - \ell')}{\ell} = 9.5 \left(\frac{70-60}{60} \right) = 9.5 \left(\frac{70}{60} - 1 \right) = \frac{9.5}{6} \Omega$$

$$F-7. \quad (a) \quad \frac{82.3}{67.3} \times 1.02 = 1.25 \text{ V}$$

(b) The high resistance to keep the initial current low when null point is being located. This saves the standard cell from damage.

(c) This high resistance does not affect the balance point because then there is no flow of current through the standard cell branch.

(d) The internal resistance of driver cell affects the current through the potentiometer wire. Since potential gradient is changed, therefore, the balance point must be affected.

(e) No, it is necessary that the emf of the driver cell is more than the emf of the cells.

(f) This circuit will not work well for measurement of small emf (mV) because the balance point will be very near to end A, and percentage error in EMF measured due to length measurement

$$\text{would be very large } E = \frac{V}{100} \ell$$

$$\Rightarrow \frac{dE}{E} = \frac{d\ell}{\ell} \text{ will be large if } \ell \text{ is very small.}$$



F-8. $\frac{x}{y} = \frac{30}{70}$ (1)

$\frac{x}{\left(\frac{12y}{12+y}\right)} = \frac{30+10}{60}$ (2)

Solving (1) & (2) $x = \frac{20}{7} \Omega$ & $Y = \frac{20}{3} \Omega$

PART - II

Section (A)

A-1. The drift velocity of electrons in a conducting wire is of the order of 1mm/s. But electric field is set up in the wire very quickly, producing a current through each cross section, almost instantaneously.

A-2. In the presence of an applied electric field ($\vec{E}$) in a metallic conductor. The electrons also move randomly but slowly drift in a direction opposite to $\vec{E}$.

A-3. $i = \frac{q}{t} = \frac{ne}{t}$

$\therefore n = \frac{i \cdot t}{e}$

Substituting $i = 3.2 \times 10^{-3} \text{ A}$
 $e = 1.6 \times 10^{-19} \text{ C}$ and $t = 1 \text{ s}$
 we get $n = 2 \times 10^{16}$

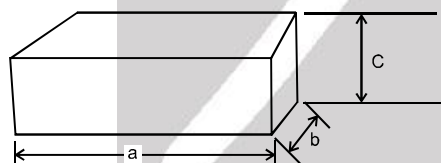
A-4 $j = \frac{I}{A}$ and $j = \frac{E}{\rho}$

$\therefore j_A > j_B$ and $E_A > E_B$.

Section (B)

B-1. Copper is metal and germanium is semiconductor. Resistance of a metal decreases and that of a semiconductor increases with decrease in temperature.

B-2.



$a > b > c$
 and $a = 2c$

$R_{\max} = \frac{\rho a}{bc}$

$R_{\min} = \frac{\rho c}{ab}$

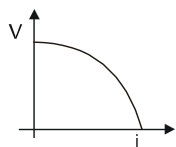
$\Rightarrow \frac{R_{\max}}{R_{\min}} = \frac{a^2}{c^2} = 4$ **Ans.**

Section (C)

C-1. In an electric circuit containing a battery, the positive charge inside the battery may go from the positive terminal to the negative terminal



C-2. Given $r \propto i \Rightarrow r = ki \quad V = E - ir = E - i(ki) \quad V = -i^2 k + E$



C-3. (a) $V = E - ir$, $V < E$ (b) $V = E + ir$, $V > E$ (C) $V = E$ (D) $V = E$

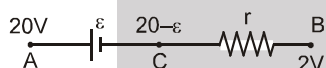
C-4. $P = \left(\frac{E}{R+5} \right)^2 R$

$\frac{dP}{dR} = 0$ at $R = 5\Omega$, so power is maximum at $R = 5\Omega$,

$R = 5\Omega \quad \frac{dP}{dR} = 0 \quad R = 5\Omega$

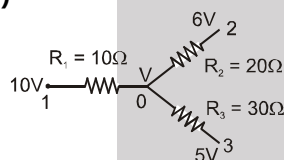
Therefore increase continuously till $R = 5\Omega$.

C-5.



Potential at C point may be greater than potential at point B. Therefore current flow in resistance may be from B to A.

C-6 (i-ii)



Apply KCL for circuit KCL

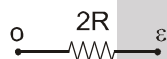
$$\frac{V-10}{10} + \frac{V-6}{20} + \frac{V-5}{30} = 0$$

$$6V - 60 + 3V - 18 + 2V - 10 = 0 \Rightarrow 11V = 88 \Rightarrow V = 8V$$

current in resistance $R_1 = \frac{10-8}{10} = 0.2$ amp

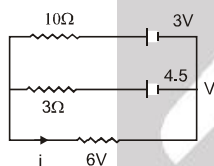
Ans

C-7.



Current flow in $2R$ resistance is from right to left.

C-8.



$$E_q = \frac{4.5}{\frac{1}{3} + \frac{1}{10}} + \frac{3}{10} = \frac{54}{13} = V, \quad r_{eq} = \frac{3 \times 10}{13} = \frac{30}{13} \Omega$$

$$i = \frac{54/13}{6 + \frac{30}{13}} = \frac{54}{108} = \frac{1}{2} \text{ amp.}$$

$$V_{6\Omega} = i.R = \frac{1}{2} \times 6 = 3V$$

Therefore current in 10Ω is zero.

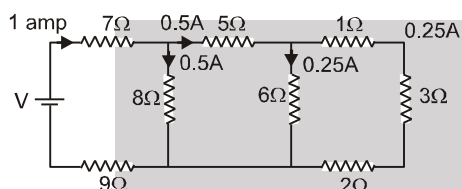


C-9. $\eta = \frac{E - Ir}{E} = 1 - \frac{r}{r+R} = \frac{R}{r+R} = 0.6 \Rightarrow R = 0.6r + 0.6R$
 $r = \frac{4}{6}R = \frac{2R}{3} \Rightarrow \eta = \frac{6R}{r+6R} = \frac{6R}{\frac{2R}{3}+6R} = \frac{18R}{2R+18R} = 0.9 = 90\%$

Section (D)

D-1. (a) $R_1 = R_{01} (1 + \alpha_1 \Delta\theta) = 600 (1 + 0.001 \times 30) = 618 \Omega$
 $R_2 = R_{02} (1 + \alpha_2 \Delta\theta) = 300 (1 + 0.004 \times 30) = 336 \Omega$
 $R_{eq} = R_1 + R_2 = 618 + 336 = 954 \Omega$

(b) $R_{eq} = R_{0eq} (1 + \alpha_{eq} \Delta\theta)$ $954 = 900 (1 + \alpha 30)$ $\alpha = \frac{54}{900 \times 30} = \frac{1}{500} \text{ degree}^{-1}$

D-2.

$$R_{eq} = 7 + 4 + 9 = 20 \Omega$$

$$V = IR_{eq} = 1 \times 20 = 20 \text{ V}$$

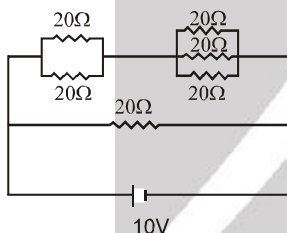
D-3. $R_{2.5W} = \frac{(110)^2}{2.5} \Omega$, $R_{100W} = \frac{(110)^2}{100} \Rightarrow R_{2.5} > R_{100}$.

In series current passes through both bulb are same

$$P_{2.5} = i^2 R_{2.5}, P_{100} = i^2 R_{100}$$

$P_{2.5} > P_{100}$ due to $R_{2.5} > R_{100}$ & $\therefore P_{2.5} > 2.5W$ & $P_{100} < 100W$ (can be verified)

Therefore 2.5 W bulb will fuse

D-4.

$$R_{eq} = \frac{\left(\frac{20}{2} + \frac{20}{3}\right) \times 20}{\frac{20}{2} + \frac{20}{3} + 20} = \frac{50 \times 20}{110}$$

$$R_{eq} = \frac{100}{11} \Omega$$

$$P = \frac{V^2}{R} = \frac{(10)^2}{100/11} = 11 \text{ W}.$$

D-5. $P = i^2 R$

current is same, so $P \propto R$

In the 1st case it is $\frac{r}{3}$, in 2nd case it is $3r$, in 3rd case it is $\frac{3r}{2}$ and in 4th case the net resistance is $\frac{2}{3}r$.

$$R_1 < R_4 < R_3 < R_2$$

$$\therefore P_1 < P_4 < P_3 < P_2$$



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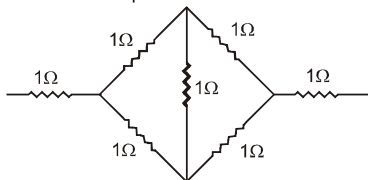
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D-6. Initial $R_{eq} = 5\Omega$



Final $R_{eq} = 3\Omega$

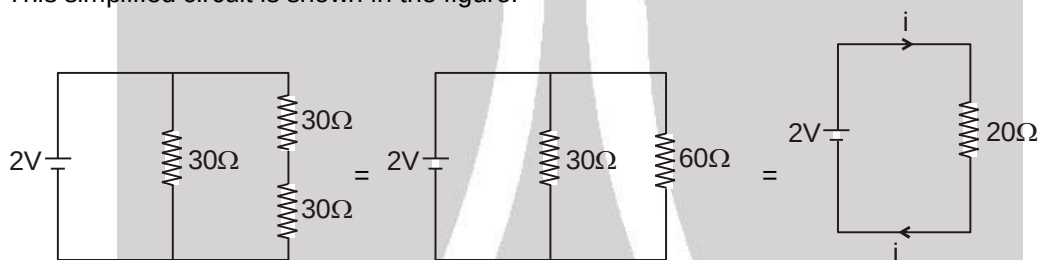
Change in resistance = $5 - 3 = 2\Omega$

D-7. $R = \frac{(220)^2}{100}$

$$R_{eq} = \frac{R}{3} + R = \frac{4}{3} R = \frac{4}{3} \frac{(220)^2}{100}$$

$$P = \frac{V^2}{R_{eq}} = \frac{(220)^2 \times 300}{4(220)^2} = \frac{300}{4} = 75 \text{ W}$$

D-8. This simplified circuit is shown in the figure.



Therefore, current $i = \frac{2}{20} = \frac{1}{10} \text{ A}$

D-9. In series, $R_{eq} = 3R$ $P = \frac{V^2}{3R} = 10 \Rightarrow \frac{V^2}{R} = 30$

in parallel $R_{eq} = R/3$

$$P = \frac{V^2}{R/3} = \frac{3V^2}{R} = 3 \times 30 = 90 \text{ W}$$

D-10. $\frac{100}{R_{eq}} = \frac{200}{40} \Rightarrow R_{eq} = 20\Omega$

$$R_{eq} = \frac{R \times 100}{100 + R} = 20\Omega \Rightarrow R = 25\Omega$$

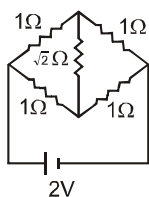
D-11. For $P_{max} \Rightarrow r = R_{eq}, R_{eq} = R/3$

$$0.1 = \frac{R}{3} \Rightarrow R = 0.3\Omega$$

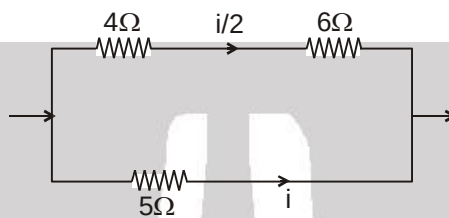


D-12. Resistance of one side = $0.1 \times 10 = 1\Omega$

$$R_{eq} = 1\Omega \quad P = \frac{V^2}{R_{eq}} = \frac{(2)^2}{1} = 4 \text{ watt}$$



D-13. Since, resistance in upper branch of the circuit is twice the resistance in lower branch. Hence, current there will be half.



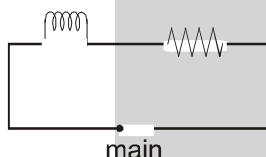
$$\text{Now } P_4 = (i/2)^2 (4)$$

$$P_5 = (i)^2 (5)$$

$$\text{or } \frac{P_4}{P_5} = \frac{1}{5}$$

$$\therefore P_4 = \frac{P_5}{5} = \frac{10}{5} = 2 \text{ cal/s.}$$

D-14.



For power across heater is maximum resistance of bulb should be minimum.

$$P_{\text{heater}} = \left(\frac{V}{R_H + R_b} \right)^2 R_H$$

R_{bulb} is minimum for 200 W.

D-15. $R_{AB} = \frac{9 \times 12}{9 + 12} + 7$

$$= \frac{85}{7} \quad \text{Ans.}$$

D-16. Condition for maximum power is

$$r = R$$

$$4 = \frac{6R \times 3R}{9 + R}$$

$$R = 2 \quad \text{Ans.}$$

D-17. $P = \frac{V^2}{r}$ $P_A = P_D = P_E > P_B = P_C$



Section (E)

E-1.
$$E_{eq} = \frac{E_1 + E_2}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} \quad r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

Therefore statement II is correct but I is wrong.

E-2. Assume M cells are connected correct and N cells connected wrong.

$$M + N = 12 \quad \dots\dots(1)$$

$$(M + 2)E - NE = 3R \Rightarrow M - N + 2 = \frac{3R}{E} \quad \dots\dots(2)$$

$$ME - (N + 2)E = 2R \Rightarrow M - N - 2 = \frac{2R}{E} \quad \dots\dots(3)$$

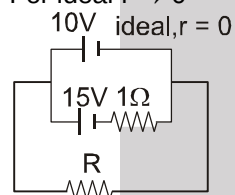
from eq (1) and (2)

$$-M + N + 10 = 0 \Rightarrow M - N = 10 \quad \dots\dots(4)$$

from eq (1) and (2)

$$M = 11, N = 1$$

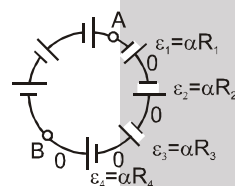
E-3. For ideal $r \rightarrow 0$



$$E = \frac{\frac{10}{r} + \frac{15}{r}}{\frac{1}{r} + \frac{1}{r}} = \frac{10+15}{1+r}$$

$$E = 10 \text{ V}$$

E-4.



$$i = \frac{\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \dots}{R_1 + R_2 + R_3 + \dots} = \frac{\alpha (R_1 + R_2 + R_3 + \dots)}{R_1 + R_2 + R_3 + \dots}$$

$i = \alpha$. so that potential difference between any two points is zero

Section (F)

F-1. $R_{eq} = 200 + \frac{300 \times 600}{300 + 600} + 100 = 500 \Omega$

$$I = \frac{100}{500} = \frac{1}{5} \text{ amp}$$

$$I_{600} = \frac{\frac{1}{5}}{\frac{1}{300} + \frac{1}{600}} \times \frac{1}{5} = \frac{1}{15} \text{ amp}$$

$$\text{Reading of volt meter} = I_{600} R_{600} = \frac{1}{15} \times 600 = 40 \text{ V}$$



F-2. Potential gradient = $\frac{E}{100}$ V/cm

$\Rightarrow$ e.m.f. of battery = $30 \times \frac{E}{100}$ **Ans.**

F-3. $R_{eq} = 2 + \frac{4}{2} + \frac{15}{3} + R_A = 9 + R_A$

$I = \frac{V}{R_{eq}} \Rightarrow 1 = \frac{10}{9 + R_A} \Rightarrow R_A = 1\Omega$

if 4Ω replace by 2Ω resistance then

$R_{eq} = 2 + \frac{2}{2} + \frac{15}{3} + 1 = 9\Omega \Rightarrow I = \frac{10}{9}$ amp

F-4 $R_{eq} = 10 + \frac{480 \times 20}{480 + 20} = 10 + \frac{96}{5} = \frac{146}{5}$

current passes through the battery.

$I = \frac{20 \times 5}{146} = \frac{100}{146} = \frac{50}{73}$ amp.

F-5. Case-I : $I_g = \frac{E_1 + E_2}{R_g + R + 2r} \Rightarrow 1 = \frac{3}{R_g + R + 2r} \Rightarrow R_g + R + 2r = 3 \dots\dots\dots (1)$

Case-II : $E_{eq} = E = 1.5$ V

$I_g = \frac{E_{eq}}{R_g + R + \frac{r}{2}} \Rightarrow 0.6 = \frac{1.5}{R_g + R + \frac{r}{2}} \Rightarrow R_g + R + \frac{r}{2} = \frac{1.5}{0.6} = 2.5 \dots\dots (2)$

from eq (1) and (2)

$\frac{3r}{2} = 0.5 \Rightarrow r = \frac{1}{3}\Omega$

F-6. $i = \frac{2}{10 + R} \quad x = \frac{V}{\ell} = \frac{2 \times 10}{(R + 10)} \cdot \frac{1}{100}$

$V_1 = x\ell \Rightarrow 10 \times 10^{-3} = \frac{2 \times 10}{(R + 10)} \times \frac{40}{100}$

$R + 10 = \frac{8}{10 \times 10^{-3}}$

$\Rightarrow R + 10 = 800 \Rightarrow R = 790\Omega$

F-7. $\frac{6}{R} = \frac{\ell}{x - \ell} \quad \frac{6}{R} = \frac{30}{20} \Rightarrow R = 4\Omega$

F-8. By ohm's law $V = iR$

$(14.4 - x) = 1.75 R$

$(22.4 - x) = 2.75 R$

$R = 8\Omega$

$14.4 - 1.75 \times 8 = x$

$x = 0.4$ volt **Ans.**

F-9. The ratio $\frac{AC}{CB}$ will remain unchanged.

F-10. As there is no change in the reading of galvanometer with switch S open or closed. It implies that bridge is balanced. Current through S is zero and $I_R = I_G, I_P = I_Q$.





PART - III

1. Drift speed $V_d = \frac{J}{ne} = \frac{i}{neA}$
 $i = \frac{V}{R}$ where $R = \frac{\rho L}{A}$ $E = \frac{V}{L}$ and $P = I^2 R$



short circuited resistor.

In a resistor current always flows from higher potential to lower potential.

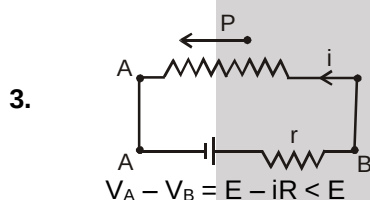
In short circuited resistor or ideal cell, energy dissipated is always zero because in short circuited resistor no current flow and in ideal cell no internal resistance.

Potential difference may be zero across a resistor, non-ideal cell or short circuited resistor.

EXERCISE-2

PART - I

1. $R \propto \frac{1}{r^4}$
 $\%R = 4 \times 0.1 = 0.4\%$
2. $V = E - Ir \Rightarrow 9.8 = 10 - I \cdot 1$
 $I = 0.2 \text{ amp}$ $R = \frac{V}{I} = \frac{9.8}{0.2} = 49 \Omega$

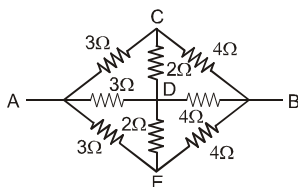


- 4.
- $V_R = \frac{E}{r+R} \Rightarrow R = \frac{E}{\frac{V_R}{r} + 1}$
- $R \rightarrow 0 \quad V_R = 0$
 & $R \rightarrow \infty \quad V_R = E$

5. For maximum power across the resistance, R is equal to equivalent resistance of remaining resistance
 $R = \frac{R_1 R_2}{R_1 + R_2}$
6. $V = \frac{E}{R + \frac{r}{2}} \times R = \frac{2ER}{2R + r}$

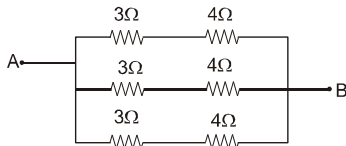


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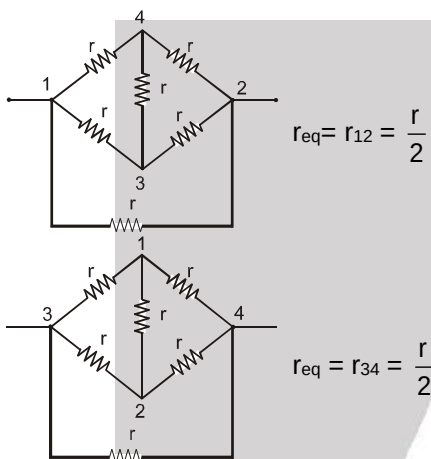


Due to input symmetry potential drop in AC, AD and AE part is same. Therefore potential at C, D and E point is same.

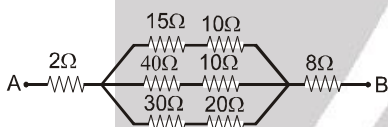
$$R_{eq} = \frac{7}{3} \Omega$$



8.

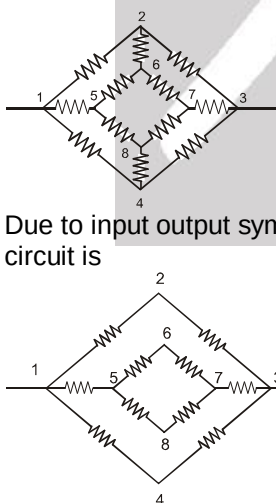


9.

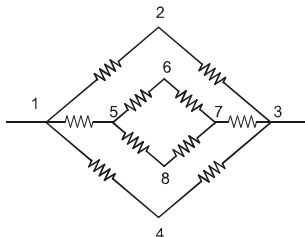


$$R_{AB} = 2 + 8 + \frac{25}{2} = \frac{45}{2} \Omega \text{ Ans}$$

10.



Due to input output symmetry, here no current passes through resistance 2 to 6 and 4 to 8. Equivalent circuit is



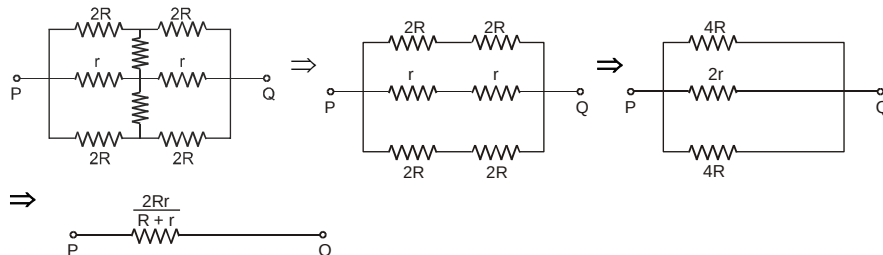
$$\frac{1}{R_{eq}} = \frac{1}{3R} + \frac{1}{2R} + \frac{1}{2R}$$

$$\frac{1}{R_{eq}} = \frac{4}{3R}$$

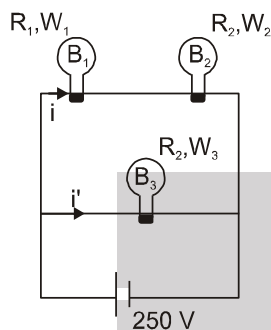




11. The circuit can be redrawn as follows :



- 12.



$$R = \frac{V^2}{P}$$

(i) $R_1 < R_2$
 $\therefore W_1 < W_2$

(ii) $R_1 + R_2 > R_2$
 $\therefore i' > i$
 $\therefore W_3 > W_2$

From (i) and (ii)

$$W_3 > W_2 > W_1$$

Alternate solution

$$P = \frac{V^2}{R}$$

$$\text{so } R = \frac{V^2}{P}$$

$$\therefore R_1 = \frac{V^2}{100}$$

$$\text{and } R_2 = R_3 = \frac{V^2}{60}$$

$$\text{Now } W_1 = \frac{(250)^2}{(R_1 + R_2)^2} \cdot R_1$$

$$W_2 = \frac{(250)^2}{(R_1 + R_2)^2} \cdot R_2$$

$$\text{and } W_3 = \frac{(250)^2}{R_3}$$

$$W_1 : W_2 : W_3 = 15 : 25 : 64 \quad \text{or } W_1 < W_2 < W_3$$

13. $(I_0 - \frac{I_0}{5}) 4 = \frac{I_0}{5} G \quad \dots(1)$

$$(I_0 - I_g) \frac{2 \times 4}{2 + 4} = I_g G \quad \dots(2)$$

from (1) and (2)

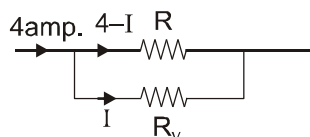
$$\frac{16}{5} I_0 \times \frac{6}{8(I_0 - I_g)} = \frac{I_0}{5 I_g}$$

$$12 I_g = I_0 - I_g \Rightarrow I_g = \frac{I_0}{13}$$

Ans



14.



$$(4 - I) R = IR_v = 20 \quad (4 - I) R = 20$$

So that, R is greater than 5Ω

15. **Case-I :** For ideal voltmeter

$$V_1 = \frac{E}{4R} \cdot 3R = \frac{3}{4} E = 0.75 E$$

Case-II :

$$V_2 = \frac{E}{7R} \cdot 6R = \frac{6}{7} E = 0.857 E$$

Case-III :

$$V_3 = \frac{E}{3R} \cdot 2R = \frac{2}{3} E = 0.666 E$$

$$16. \quad 0.75 \times 60 = 15 \times i_2 \quad i_2 = 3 \text{ amp}$$

$$17. \quad i_1 = 0.75 + 3 \quad (\text{form KCL})$$

$$i_1 = 3.75 \text{ amp}$$

18. For $I_{\max}$, R_h is minimum which is zero.

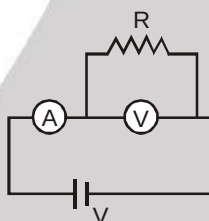
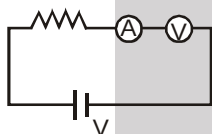
$$I_{\max} = \frac{5.5}{20} \text{ Amp.}$$

for $I_{\min}$, R_h is maximum which is 30Ω .

$$I_{\min} = \frac{5.5}{20+30} = \frac{5.5}{50} \text{ Amp.}$$

$$\frac{I_{\min}}{I_{\max}} = \frac{5.5}{50} \times \frac{20}{5.5} = \frac{2}{5} \text{ Amp.} \quad \text{Ans}$$

19.

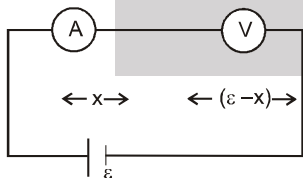


Equivalent resistance decrease so current will increase.

$$V_A + V_V = V$$

Due to change, V_A increases so voltmeter reading will decrease.

20.



$$\frac{(\epsilon - x)}{\eta} + \eta x = \epsilon \Rightarrow x = \frac{\epsilon}{\eta + 1}$$

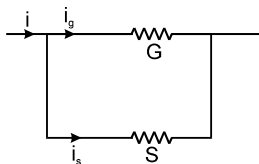
Reading of voltmeter after connection of resistance is $\frac{\epsilon - x}{\eta}$

$$= \frac{\epsilon}{(\eta + 1)} = \frac{6}{2 + 1} = 2V$$

Ans



21.



$$i = i_g + i_s \quad \dots\dots\dots(1)$$

$$i_g G = i_s S \quad \dots\dots\dots(2)$$

from (1) & (2) (putting $i_g = 0.1 \text{ mA}$, $G = 100\Omega$) we have
 $i = 100.1 \text{ mA}$

22. BC, CD and BA are known resistance,
 The unknown resistance is connected between A and D.
 Hence, the correct option is (D)

23. $\frac{RR_v}{R+R_v} \cdot I = V$

$$\frac{RR_v}{R+R_v} \frac{V}{I} \Rightarrow \frac{I}{V} = \frac{1}{R} + \frac{1}{R_v}$$

$$\Rightarrow \frac{I}{V} - \frac{1}{R_v} = \frac{1}{R} \Rightarrow R = \frac{1}{\frac{I}{V} - \frac{1}{R_v}} = \frac{V}{I - \frac{V}{R_v}}$$

24. $R = 10\Omega$

$$P_{av} = \frac{\frac{(4)^2}{10} \times 0.1 + \frac{(2)^2}{10} \times 10.1}{0.2} = \frac{1.6 + 0.4}{2} = 1 \text{ watt}$$

25. $\epsilon_1 - \left\{ \frac{\epsilon_1 + \epsilon_2}{r_1 + r_2 + R} \right\} r_1 = 0$

$$r_1 + r_2 + R = 2r_1 \Rightarrow R = r_1 - r_2$$

26. $\frac{P}{Q} = \frac{R}{X} \Rightarrow \frac{10}{100} = \frac{600}{X}$

$$X = 6000 \Omega$$

For second case

$$\frac{P}{Q} = \frac{R}{X} \Rightarrow \frac{10}{100} = \frac{630}{X} \Rightarrow X = 6300 \Omega$$

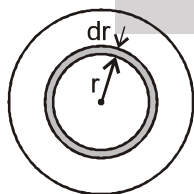
$$R_f = R_0 (1 + \alpha \Delta T)$$

$$6300 = 6000 (1 + \alpha (100))$$

$$\alpha = 5 \times 10^{-4} / ^\circ\text{C}$$

PART - II

1.



$$(a) J = J_0 \left(1 - \frac{r}{R} \right) \quad \text{shaded area } dA = 2\pi r dr \quad di = 2\pi J_0 \left(r - \frac{r^2}{R} \right) dr$$

$$i = 2\pi J_0 \int_0^R \left(r - \frac{r^2}{R} \right) dr = 2\pi J_0 \left[\frac{R^2}{2} - \frac{R^2}{3} \right] = \frac{2\pi J_0 R^2}{6} = \frac{J_0 A}{3} \quad \text{Put } A = 4$$



2. Let length of P = ℓ meter Resistance of P = $\rho \frac{\ell}{A}$.

Let length of Q = $(1 - \ell)$ meter

$$\text{Resistance of R} = \rho \frac{\frac{2\ell}{2}}{A} = 4\rho \frac{\ell}{A}$$

$$\text{Resistance of Q} = \rho \frac{(1-\ell)}{A} = 4\rho \frac{\ell}{A} \Rightarrow 4\ell = 1 - \ell \quad \ell = \frac{1}{5} \quad 1 - \ell = \frac{4}{5}$$

$$\frac{\text{Resistance of P}}{\text{Resistance of Q}} = \frac{\rho \frac{\ell}{A}}{\rho \frac{(1-\ell)}{A}} = \frac{\ell}{1-\ell} = \frac{\frac{1}{5}}{\frac{4}{5}} = \frac{1}{4} \quad \text{Ans.}$$

$$\frac{\text{Length of P}}{\text{Length of Q}} = \frac{\ell}{1-\ell} = \frac{1}{4} \quad \text{Ans.}$$

3. For maximum power $r_{eq} = R_{eq} \Rightarrow 2 + \frac{6x}{6+x} = 4 \Rightarrow \frac{12+8x}{6+x} = 4$
 $\Rightarrow 12 + 8x = 24 + 4x \Rightarrow 4x = 12 \quad x = 3\Omega \quad \text{Ans.}$

4. Condition for maximum power is
 Net internal resistance = Net external resistance

$$r_{net} = R_{net}$$

$$\frac{300}{n} \times r = R$$

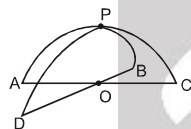
$$n^2 = \frac{300 \times r}{R}$$

$$n = \sqrt{\frac{300 \times 0.3}{10}}$$

$$n = 3$$

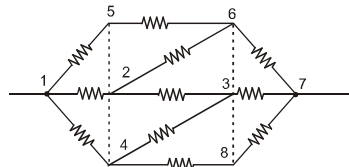
5. (a) $E = 6V, r = 10\Omega, \quad r_1 = 1\Omega$
 $9V = 6 + i \times 10 \Rightarrow i = \frac{3}{10}$
 $i = 0.3A \quad \text{Ans.}$
- (b) When internal resistance $r_1 = 1\Omega$
 $9 = 6 + i_1 \times 1 \Rightarrow i_1 = 3A \quad \text{Ans.}$

6.

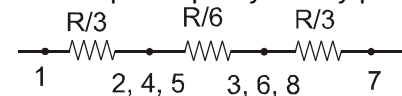


$$R_{eff} = \frac{1}{4} \left[\frac{\pi a}{2} r + ar \right] = \frac{1}{4} ar \left[\frac{\pi + 2}{2} \right] = \left[\frac{\pi + 2}{8} \right] ar.$$

7.



due to input output symmetry potential at point 2, 4, 5, are equal and potential at point 3, 6, 8 are equal

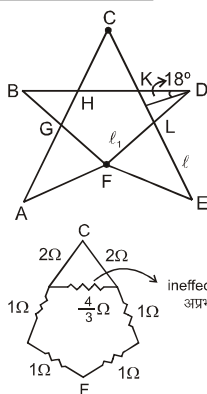


$$R_{eq} = \frac{R}{3} + \frac{R}{6} + \frac{R}{3} = \frac{5}{6} R$$





8.



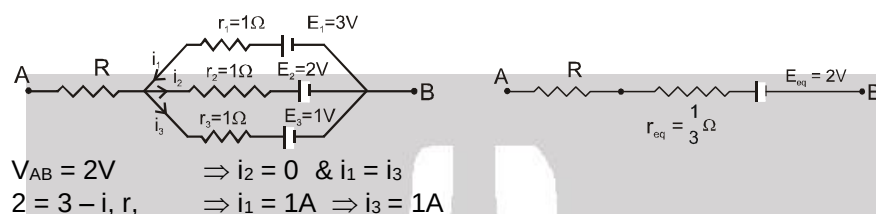
$$\ell_1 = 2\ell \sin 18^\circ = \frac{2\ell}{3}$$

$$R_{LK} = 2 \times \frac{2}{3} = \frac{4}{3} \Omega$$

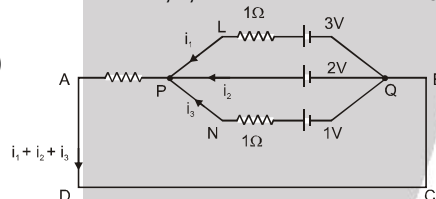
$$R_{FC} = \frac{4}{2} = 2 \Omega$$

9.

(i)



(ii)

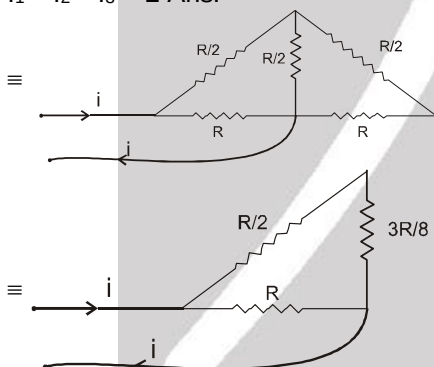


Apply kvl in ABCD

$$2 - (i_1 + i_2 + i_3)R = 0$$

$$i_1 + i_2 + i_3 = 2 \text{ Ans.}$$

10.



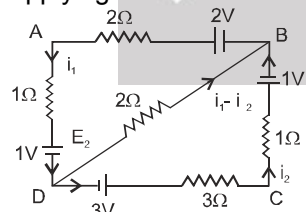
(by folding)

$$R_{\text{eff}} = \frac{\frac{7R}{8} \times R}{\frac{7R}{8} + R} = \frac{7R}{15}$$

$$i = \frac{U}{R_{\text{eff}}} = \frac{15U}{7R}$$

11.

Applying Kirchhoff's second law in loop ADBA :



$$2 - 2i_1 - i_1 - 1 - 2(i_1 - i_2) = 0 \quad \dots\dots\dots(1)$$

Similarly applying Kirchhoff's second law in loop BDCB

$$2(i_1 - i_2) + 3 - 3i_2 - i_2 - 1 = 0 \quad \dots\dots\dots(2)$$

Solving Equation (1) and (2) we get,

$$i_1 = \frac{5}{13}, i_2 = \frac{6}{13} \quad \text{and} \quad i_1 - i_2 = -\frac{1}{13}$$



- (i) Potential difference between B and D.

$$V_B + 2(i_1 - i_2) = V_D$$

$$\therefore V_B - V_D = -2(i_1 - i_2) = \frac{2}{13} \text{ volt}$$

- (ii)
- $V_G = E_G - i_2 r_G = 3 - \frac{6}{13} \times 3 = \frac{21}{13} \text{ volt}$

$$V_H = E_H + i_2 r_H = 1 + \frac{6}{13} \times 1 = \frac{19}{13} \text{ volt}$$

$$12. \quad 2 = \frac{12}{10000 + R} R$$

$$\Rightarrow R = 2000 \Omega = 2 \text{ K}\Omega,$$

Galvanometer will show deflection, as the temperature of wire wound decreases, resistance decreases.

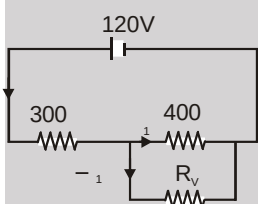
- 13.
- $E = 1.52 \text{ V} \quad V = E - ir$

$$1.45 = 1.52 - 1r$$

$$r = 0.07 \Omega$$

Ans.

- 14.
- Case-I**



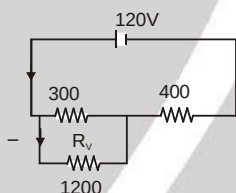
$$\text{Current } I = \frac{60}{300} = \frac{1}{5} \text{ amp}$$

$$I_1 = \frac{60}{400} = \frac{3}{20} \text{ amp.}$$

$$60 = (I - I_1) R_v$$

$$\Rightarrow R_v = \frac{60}{\frac{1}{5} - \frac{3}{20}} = 1200 \Omega$$

- Case-II**



$$I = \frac{120}{400 + \frac{300 \times 1200}{1200 + 300}} = \frac{6}{32} \text{ amp.}$$

$$I_0 \cdot 300 = (I - I_0) \cdot 1200 \Rightarrow I_0 = \frac{1200}{1500} \quad I = \frac{4}{5} \times \frac{3}{16} = \frac{3}{20} \text{ amp}$$

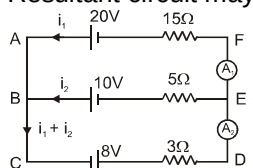
$$\text{Reading of voltmeter} = \frac{3}{20} \times 300 = \frac{900}{20} = 45 \text{ V}$$

Ans

15. As the ammeters
- A_1
- and
- A_2
- are ideal, potential drop across AB and AC are zero. Hence point B and C are at equal potential, so there will be no current through
- A_3
- .

$$I_3 = 0$$

Resultant circuit may be drawn as



Applying KVL in the loop ABEFA

$$-10 + i_2 \cdot 5 - 15 i_1 + 20 = 0 \Rightarrow 3i_1 - i_2 = 2 \quad \dots(1)$$

Applying KVL in the loop BCDEB



$$8 - (i_1 + i_2) 3 - i_2 5 + 10 = 0 \Rightarrow 3i_1 + 8i_2 = 18 \quad \dots(2)$$

$$i_2 = \frac{16}{9} \text{ Amp}, i_1 = \frac{34}{27} \text{ Amp}$$

Reading of ammeter A_1 ,

$$i_1 + i_2 = \frac{82}{27} \text{ amp}$$

Ans.

Reading of ammeter A_2 ,

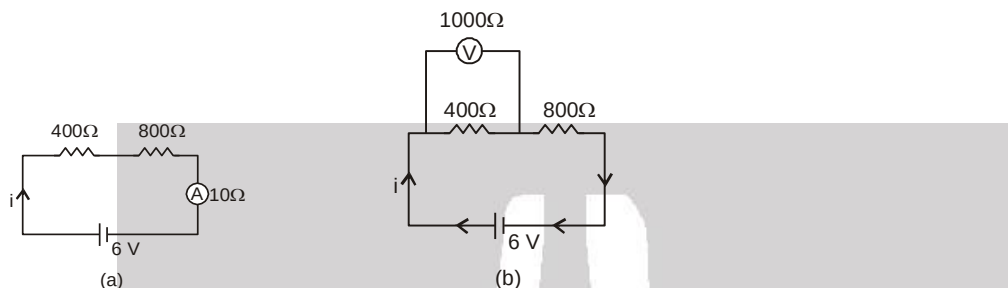
$$i_1 = \frac{34}{27} \text{ amp}$$

Ans.

$$i_{A1} + i_{A2} + i_{A3}$$

$$\frac{82}{27} + \frac{34}{27} + 0 = \frac{116}{27} = 58 \left(\frac{2}{27} \right)$$

16.



Refer figure (a) : Current through ammeter,

$$i = \frac{\text{net emf}}{\text{net resistance}} = \frac{6}{400 + 800 + 10} = 4.96 \times 10^{-3} \text{ A} = 4.96 \text{ mA}$$

Refer figure (b) : Combined resistance of 1000Ω voltmeter and 400Ω resistance is,

$$R = \frac{1000 \times 400}{1000 + 400} = 285.71 \Omega$$

$$\therefore i = \frac{6}{(285.71 + 800)} = 5.53 \times 10^{-3} \text{ A}$$

Reading of voltmeter

$$= V_{ab} = iR = (5.53 \times 10^{-3}) (285.71) = 1.58 \text{ volt}$$

PART - III

- Electrons are accelerated in opposite direction of electric field. Therefore speed of the electron is more at B than at A.
- In series current remain same $I = neAv_d$, $J = I/A$, for constant current $v_d \propto 1/A$ and $J \propto 1/A$.
- When no current is passed through a conductor the average velocity of a free electron over a large period of time is zero and the average of the velocities of all the free electrons at an instant is zero.
- $IR = V = E\ell$

$$\Rightarrow I \frac{\rho \ell}{A} = E\ell$$

$$\Rightarrow \rho = \frac{EA}{I} = \frac{E}{J} = \frac{5 \times 10^{-2}}{10} = 5 \times 10^{-3} \Omega\text{-m}$$

$$\Rightarrow \sigma = \frac{1}{\rho} = \frac{1}{5 \times 10^{-3}} = 200 \text{ mho/m.}$$

$$5. \quad (i) R_{\text{bulb}} = \frac{V}{I} = \frac{10}{10 \times 10^{-3}} = 1. \text{ k}\Omega$$

$$(ii) R_{\text{bulb}} = \frac{220}{50 \times 10^{-3}} = 4.4 \text{ k}\Omega.$$

since increase in temperature increases resistance when it is connected to 220 V mains.



6. (A) $dR = \frac{x}{\sigma_0} \frac{dx}{\ell A}$, $R = \int_0^\ell \frac{x}{\sigma_0} \frac{dx}{\ell A} = \frac{\ell}{2\sigma_0 A}$

(B) $I = \frac{V_0}{R}$

(C) $J = \frac{I}{A}$

(D) $J = \sigma E$

$E = \frac{J}{\sigma} = \frac{2V_0}{\ell^2} x$

7. $E_{eq} = \frac{\frac{KE}{r} + \frac{KE}{r} + \frac{KE}{r} + \dots \text{upto } \frac{N}{K}}{\frac{1}{r} + \frac{1}{r} + \frac{1}{r} + \dots \text{upto } \frac{N}{K}}$

$E_{eq} = KE$, $\frac{1}{r_{eq}} = \frac{1}{Kr} + \frac{1}{Kr} \dots \text{upto } \frac{N}{K}$ $r_{eq} = \frac{K^2 r}{N}$

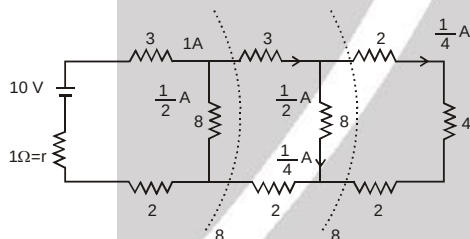
For maximum power $r_{eq} = R \Rightarrow \frac{K^2 r}{N} = R \Rightarrow K = \sqrt{\frac{NR}{r}}$

Maximum power = $P_{max} = I^2 R = \left(\frac{E_{eq}}{2R}\right)^2 R = \left(\frac{EK}{2R}\right)^2 R = \frac{E^2 K^2}{4R} = \frac{E^2 \times \left(\frac{NR}{r}\right)}{4R} = \frac{NE^2}{4r}$

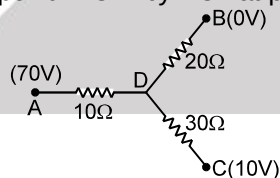
8*. Equivalent resistance $R_{eq} = 10 \Omega$ so current passing through battery and 3Ω resistance is

$i = \frac{10}{10} = 1 \text{ A}$ **Ans.**

and current passing through 4Ω is 0.25 A **Ans.**



9*. Let potential of point D is x . by KCL at point D.



$I_1 + I_2 + I_3 = 0$

$\frac{x-70}{10} + \frac{x-0}{20} + \frac{x-10}{30} = 0$

$\Rightarrow 6x - 420 + 3x + 2x - 20 = 0$

$\Rightarrow 11x = 440 \Rightarrow x = 40 \text{ volt}$

$\therefore I_1 = \frac{40-70}{10} = -3\text{A}, I_2 = \frac{40}{20} = 2\text{A}$



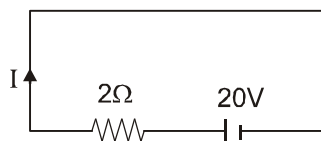
$$I_3 = \frac{40 - 10}{30} = 1A$$

$$P = i^2 R$$

$$P = 3^2 \times 10 + 2^2 \times 20 + 1^2 \times 30$$

$$P = 200 \text{ W}$$

10.*



Current flow in circuit is $I = 10 \text{ amp}$

Power supplied by the battery is $= I^2 R = (10)^2 \times 2 = 200 \text{ W}$

Potential drop across 4Ω & 6Ω are equal and it is equal to zero.

Current in AB wire is 10 amp .

11*.

$$V = E - ir$$

from graph $V = 10 - 5i \Rightarrow r = 5\Omega, E = 10V$

$$i_{\max} = \frac{E}{r} = \frac{10}{5} = 2 \text{ amp}$$

12*.

$$\text{for short circuited, } I = \frac{E}{r}$$

$$V = E - Ir = E - \frac{E}{r} \cdot r = 0$$

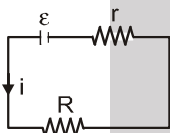
When current flow from negative terminal to positive terminal

$$V = E - Ir \text{ which is less than } E$$

When current flow from positive terminal to negative terminal

$$V = E + Ir \text{ which is greater than } E.$$

13*.



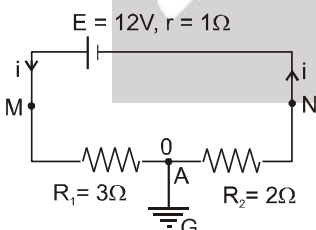
$$\text{Current } i = \frac{\varepsilon}{R + r}$$

Cell generating power $= \varepsilon i$

$$\text{Heat produced in } R \text{ at the rate} = i^2 R = iR \cdot \frac{\varepsilon}{R + r} = \varepsilon i \cdot \frac{R}{R + r}$$

$$\text{Heat produced in } r \text{ at the rate} = i^2 r = \varepsilon i \cdot \frac{r}{R + r}$$

14*.



Current in the circuit

$$i = \frac{12}{3 + 2 + 1} = 2 \text{ amp}$$

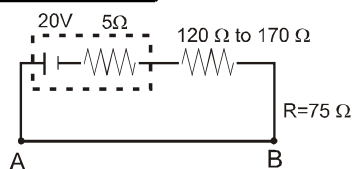
$$V_A = 0 \Rightarrow V_A - V_M = -2 \times 3 \Rightarrow V_M = 6V \text{ and}$$

$$V_N - V_A = -2 \times 2 \Rightarrow V_N = -4V$$

Current in wire AG is zero.



15.



$$i_{\max} = \frac{20}{5 + 75 + 120} = \frac{1}{10} \text{ amp}$$

$$V_{\max} = i_{\max} R_{75} = \frac{1}{10} \times 75 = 7.5 \text{ V}$$

Range of potentiometer 0 to 7.5 V

16.

For non ideal ammeter and voltmeter, ammeter have low resistance and voltmeter have high resistance. Therefore the main current in the circuit will be very low and almost all current will flow through the ammeter. If emf of cell is very high then current in ammeter is very high result of this current the devices may get damaged. If devices are ideal that means resistance of voltmeter is infinity. so that current in the circuit is zero. Therefore ammeter will read zero reading and voltmeter will read the emf of cell.

17.

$$\text{For } 50 \text{ V, } R_v = \frac{50}{50 \times 10^{-6}} = 1000 \text{ K}\Omega \text{ in series}$$

$$\text{For } 10 \text{ V, } R_v = \frac{10}{50 \times 10^{-6}} = 200 \text{ K}\Omega \text{ in series}$$

$$\text{For } 5 \text{ mA, } R_s = \frac{100 \times 50 \times 10^{-6}}{5 \times 10^{-3}} = 1\Omega \text{ in parallel}$$

$$\text{For } 10 \text{ mA, } R_s = \frac{100 \times 50 \times 10^{-6}}{10 \times 10^{-3}} = \frac{1}{2} \Omega \text{ in parallel}$$

18.

$$V_A - V_B = E_{\text{eq}} = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

If $\varepsilon_1 > \varepsilon_2$ source ε_1 act as a source and ε_2 act as a load.

$$\text{and } V_1 = \varepsilon_1 - ir_1$$

$$V_2 = \varepsilon_2 + ir_2$$

$$V_1 = V_2 \text{ as } i = \frac{\varepsilon_1 - \varepsilon_2}{r_1 + r_2} \quad \text{for } \varepsilon_1 > \varepsilon_2$$

19*.

In parallel combination potential difference are same

$$\text{So } V_1 + V_2 = V_3$$

Here V_1 and V_2 are in series but their resistances are different

$$\text{So } V_1 = iR_1$$

$$V_2 = iR_2$$

$$V_1 \neq V_2$$

20.

$$\text{Potential gradient } x = \frac{E}{r + R} \times \frac{R}{100}$$

Where R = resistance of potentiometer wire.

$$\frac{E}{2} = x\ell \Rightarrow \frac{E}{2} = \frac{E}{r + R} \times \frac{R}{100} \ell$$

$$\ell = \frac{50(r + R)}{R} \Rightarrow \ell > 50 \text{ cm.} \quad \text{Ans.}$$

 $\therefore$ Balance length should be less than or equal to 100 cm

$$\ell \leq 100$$

$$\Rightarrow \frac{50(r + R)}{R} \leq 100 \Rightarrow R \geq r \quad \text{Ans.}$$



21. It is easier to start a car engine on a warm day than on a chilly cold day because the internal resistance of battery decreases with rise in temperature and so current increases.

$$\text{Power Loss} = I^2 R, \quad \Rightarrow \quad \text{Power loss} \propto I^2$$

$$\text{Also } P = V \cdot I \quad \Rightarrow \quad I = \frac{P}{V}$$

Since for given power & line P & R are constant

$$\text{Power loss} = I^2 R = \frac{P^2 R}{V^2} \quad \therefore \quad \text{Power loss} \propto \frac{1}{V^2}$$

mica is good conductor of heat but bad conductor of electricity

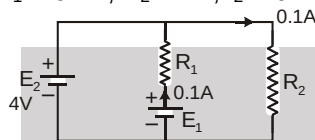
PART - IV

1 - 3.

As E_2 is increasing it's current also increases, So, increasing graph is of i_2 .

$$i_1 = 0.1 \text{ A}, E_2 = 4 \text{ V}, i_2 = 0$$

As ;

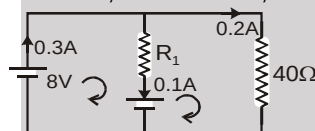


$$0.1 R_1 + 0.1 R_2 - E_1 = 0$$

$$0.1 R_2 - 4 \text{ V} = 0$$

$$R_2 = 40 \Omega$$

Now ; $i_2 = 0.3 \text{ A}, i_1 = -0.1 \text{ A}, E_2 = 8 \text{ V}$



$$\text{Now ; } 0.1 R_1 + E_1 - 8 = 0$$

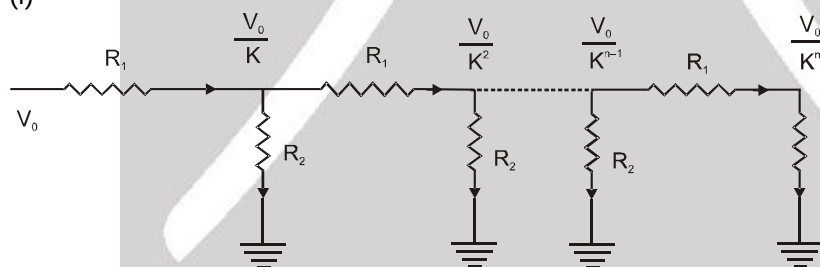
When $E_2 = 6 \text{ V}$, current in E_1 is $i_1 = 0$ (from graph)

$$E_1 = 6 \text{ V}$$

$$\Rightarrow R_1 = \frac{4}{0.2} = 20 \Omega$$

5.

(i)



Energy kirchoffs first law at first junction

$$\frac{V_0 - \frac{V_0}{K}}{R_1} = \frac{\frac{V_0}{K} - \frac{V_0}{K^2}}{R_1} + \frac{\frac{V_0}{K} - 0}{R_2} \Rightarrow \left(1 - \frac{1}{K}\right) \frac{V_0}{R_1} = \frac{V_0}{KR_1} \left(1 - \frac{1}{K}\right) + \frac{V_0}{KR_2}$$

$$\frac{1}{R_1} \left(1 - \frac{1}{K} - \frac{1}{K} + \frac{1}{K^2}\right) = \frac{1}{KR_2} \Rightarrow \frac{K^2 - 2K + 1}{K^2} = \frac{R_1}{KR_2}$$

$$\frac{R_2}{R_1} = \frac{K}{(K-1)^2} \quad \dots\dots\dots(1)$$

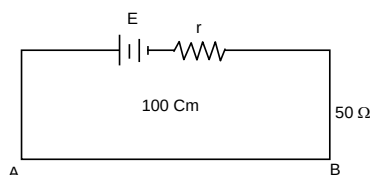
$$\frac{\frac{V_0}{K^{n-1}} - \frac{V_0}{K^n}}{R_1} = \frac{V_0}{KR_2}, \quad \frac{R_1}{R_3} = (K-1) \quad \dots\dots\dots(2)$$



$$(1) \& (2) \Rightarrow \frac{R_2}{R_3} = \frac{K}{K-1}$$

$$(ii) I = \frac{V_1}{R_2} = \frac{V_0/K}{R_3(K/K-1)} = \frac{V_0}{K^2 R_3} (K-1) \Rightarrow I = \left(\frac{K-1}{K^2} \right) \frac{V_0}{R_3}$$

6.



$$\text{Current in AB} = \frac{E}{50 + 4}$$

Potential gradient

$$x = \left(\frac{E}{50 + 4} \right) \frac{50}{100}$$

For balancing length

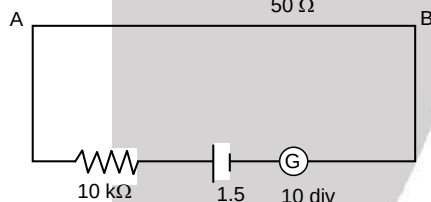
$$v = x\ell$$

$$1.5 = \left(\frac{E}{50 + 4} \right) \frac{1}{2} \times 40$$

$$E = 4.05$$

50 Ω

7.



$$R_G = 50 \Omega$$

$$I = \frac{1.5}{10100}$$

$$\text{Current sensitivity} = \frac{\text{Deflection}}{\text{Current}}$$

$$\frac{10}{1.5} \times 10100$$

$$\frac{10 \times 0.010100}{1.5} = 0.067 \frac{\text{Div}}{\mu\text{A}}$$

8.

Current through G is Case – I

$$I_0 = \frac{E}{R + G}$$

Current through G is Case – II

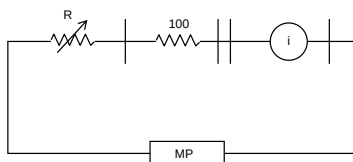
$$I_1 = \frac{E}{\left(\frac{GS}{S+G} + R \right)} \times \frac{S}{(S+G)}$$

$$R \gg G \quad I_1 = \frac{I_0}{2}$$





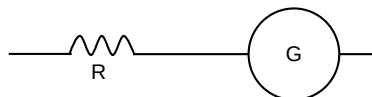
9 to 12.

Case-1

$$10 i_0 G = 12 \times 10^{-3}$$

$$10 i_0 (100) = 16 \times 10^{-3} \Rightarrow 10^3 i_0 = 16 \times 10^{-3} \Rightarrow i_0 = 16 \times 10^{-6} \text{ A}$$

$$\frac{G}{100} = \frac{3}{4} \Rightarrow G = 75 \Omega$$

Case-2

$$30 i_0 (R + G) = 12$$

$$R + G = \frac{12}{30 i_0} = \frac{2}{5 \times 16 \times 10^{-6}} = \frac{2 \times 10^6}{80} = 2.5 \times 10^4 \Omega = 25 \text{ k}\Omega$$

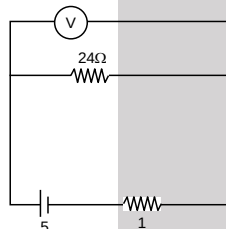
$$R = 25 \text{ k}\Omega - 75 \Omega$$

Case-3

$$30 i_0 (R + G) = 12$$

$$n i_0 (R + G) = \left(\frac{5}{25}\right) 24$$

$$n = 12$$



EXERCISE-3

PART - I

1. $R = \frac{\rho l}{A}$
 $R = \frac{\rho L}{tL} = \frac{\rho}{t}$
 Independent of L.

2. $100 = \frac{V^2}{R'_{100}} \Rightarrow \frac{1}{R'_{100}} = \frac{100}{V^2}$

where R'_{100} is resistance at any temperature corresponds to 100 W

$$60 = \frac{V^2}{R'_{60}} \Rightarrow \frac{1}{R'_{60}} = \frac{60}{V^2}$$

$$40 = \frac{V^2}{R'_{40}} \Rightarrow \frac{1}{R'_{40}} = \frac{40}{V^2}$$

From above equations we can say

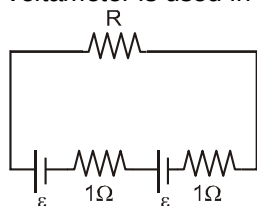
$$\frac{1}{R'_{100}} > \frac{1}{R'_{60}} > \frac{1}{R'_{40}}$$

So, most appropriate answer is option (D).



3. To verify Ohm's law one galvanometer is used as ammeter and other galvanometer is used as voltmeter. Voltmeter should have high resistance and ammeter should have low resistance as voltmeter is used in parallel and ammeter in series that is in option (C).

4.



$$i = \frac{2\varepsilon}{2+R}$$

$$J_1 = \left(\frac{2\varepsilon}{2+R} \right)^2 R$$

$$\varepsilon_{eq} = \frac{\frac{\varepsilon}{1} + \frac{\varepsilon}{1}}{\frac{1}{1} + \frac{1}{1}} = \varepsilon$$

$$r_{eq} = \frac{1}{2} \Rightarrow i = \frac{\varepsilon}{\frac{1}{2} + R} = \frac{2\varepsilon}{2R+1}$$

$$J_2 = \left(\frac{2\varepsilon}{1+2R} \right)^2 R$$

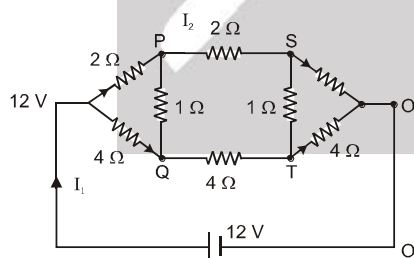
$$\text{Given } J_1 = \frac{9}{4} J_2$$

$$\Rightarrow \left(\frac{2\varepsilon}{2+R} \right)^2 R = \frac{9}{4} \left(\frac{2\varepsilon}{1+2R} \right)^2 R \Rightarrow \frac{2}{2+R} = \frac{3}{1+2R}$$

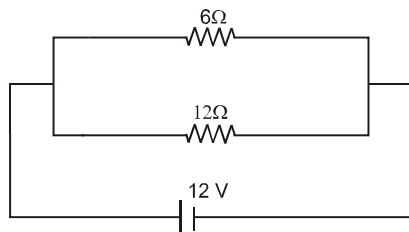
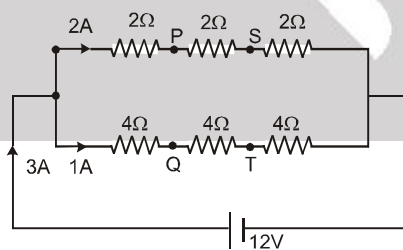
$$\Rightarrow 2 + 4R = 6 + 3R \Rightarrow R = 4\Omega.$$

5. $\varepsilon = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{6}{1} + \frac{3}{2}}{\frac{1}{1} + \frac{1}{2}} = \frac{15}{3} = 5 \text{ volt}$ **Ans.**

6. Due to input and output symmetry P and Q and S and T have same potential.



$\Rightarrow$





$$R_{eq} = \frac{6 \times 12}{18} = 4\Omega$$

$$I_1 = \frac{12}{4} = 3A$$

$$I_2 = \left(\frac{12}{6+12} \right) \times 3$$

$$I_2 = 2A$$

$$V_A - V_S = 2 \times 4 = 8V$$

$$V_A - V_T = 1 \times 8 = 8V$$

$$V_P = V_Q \Rightarrow \text{Current through PQ} = 0 \quad (A)$$

$$V_P = V_Q \Rightarrow V_Q > V_S \quad (C)$$

$$I_1 = 3A \quad (B)$$

$$I_2 = 2A \quad (D)$$

7. In given Kettle $R = \rho \frac{L}{\pi \left(\frac{d}{2} \right)^2} = \frac{4\rho L}{\pi d^2}$, $P = \frac{V^2}{R}$

In second Kettle $R_1 = \rho \frac{L}{\pi d^2}$ $R_2 = \frac{\rho L}{\pi d^2}$

So $R_1 = R_2 = R/4$

If wires are in parallel equivalent resistance

$$R_P = \frac{\rho L}{\pi d^2}$$

Then power $P_P = 8P$

So it will take 0.5 minute

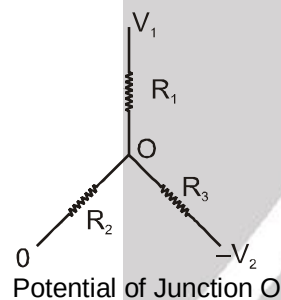
If wires are in series equivalent resistance

$$R_S = R/2$$

Then power $P_S = 2P$

So it will take 2 minutes

8.



Potential of Junction O

$$V_0 = \frac{\frac{V_1}{R_1} + 0 - \frac{V_2}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Current through R_2 will be zero if

$$V_0 = 0 \Rightarrow \frac{V_1}{V_2} = \frac{R_1}{R_3}$$

9. $\frac{6}{1000} (G + 4990) = 30$

$$\Rightarrow G + 4990 = \frac{30,000}{6} = 5000$$

$$\Rightarrow G = 10$$



$$\frac{6}{1000} \times 10 = \left(1.5 - \frac{6}{1000}\right) S$$

$$\Rightarrow S = \frac{60}{1494} = \frac{2n}{249}$$

$$\Rightarrow n = \frac{249 \times 30}{1494} = \frac{2490}{498} = 5$$

10. For balanced meter bridge

$$\frac{X}{R} = \frac{\ell}{(100 - \ell)}, \quad \frac{X}{40} = \frac{90}{60} \Rightarrow X = 60\Omega$$

$$X = R \frac{\ell}{(100 - \ell)}$$

$$\frac{\Delta X}{X} = \frac{\Delta \ell}{\ell} + \frac{\Delta \ell}{100 - \ell} = \frac{0.1}{40} + \frac{0.1}{60}$$

$$\Delta X = 0.25$$

$$\text{so } X = (60 \pm 0.25) \Omega$$

11. R_{Al} and R_{Fe} are Parallel to each other

$$\frac{1}{R_{eq}} = \frac{1}{R_{Al}} + \frac{1}{R_{Fe}}$$

$$\frac{1}{R_{Al}} = \frac{2.7 \times 10^{-8} \times (.05)}{(.007)^2 - (.002)^2};$$

$$R_{Al} = \frac{2.7 \times 10^{-8} \times 5 \times 10^{-4}}{10^{-6} \times 45 \times 100} = 3 \times 10^{-5}$$

$$R_{Fe} = \frac{1.0 \times 10^{-7} \times (.05)}{(.002)^2}$$

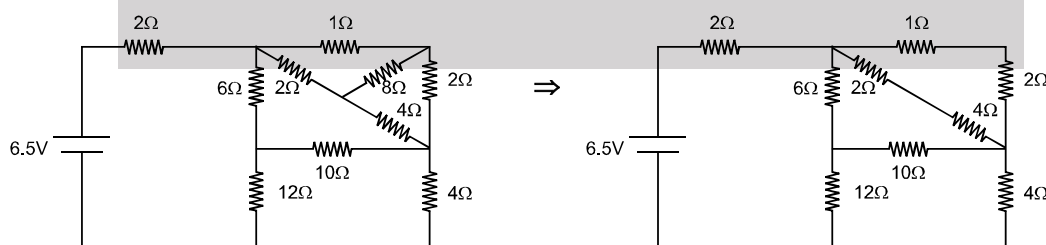
$$R_{Fe} = \frac{10^{-7}}{4 \times 10^{-6}} 5 \times 10^{-2} = \frac{5}{4} \times 10^{-3}$$

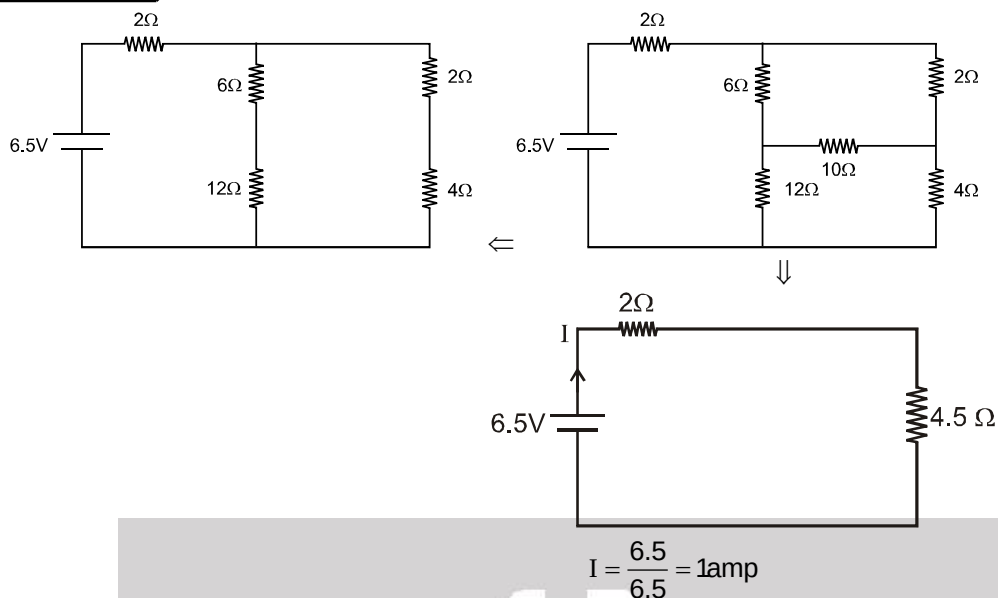
$$\frac{1}{R_{eq}} = \frac{1}{3 \times 10^{-5}} + \frac{4}{5 \times 10^{-3}} = \frac{10^5}{3} + \frac{4 \times 10^3}{5}$$

$$\frac{1}{R_{eq}} = \frac{15}{5 \times 10^5 + 12 \times 10^3} = \frac{15}{5.12 \times 10^5}$$

$$\frac{1}{R_{eq}} = \frac{150}{5.12} \mu\Omega = \frac{1875}{64} \mu\Omega$$

12.

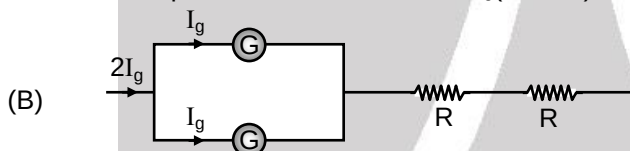




13. Suppose current for full deflection (i.e., For maximum range) through galvanometer is I_g .



Total potential difference $V_1 = 2I_g(R + R_c)$



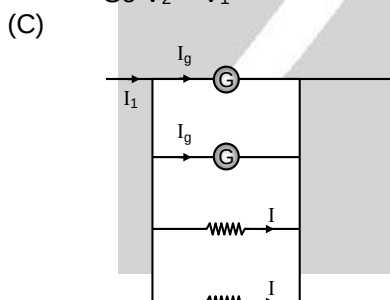
Total potential difference $V_2 = 2I_g \left(\frac{R_c}{2} + 2R \right) = I_g(R_c + 4R)$

Now since $2R_c < R$

So $V_1 < 3I_g R$

while $V_2 > 4RI_g$

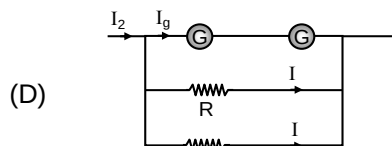
So $V_2 > V_1$



$$I_g R_c = IR$$

$$I = I_g \frac{R_c}{R}$$

$$I_1 = 2I_g \left(1 + \frac{R_c}{R} \right)$$

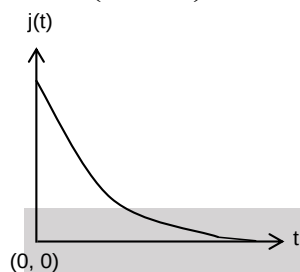


$$2I_g R_c = IR$$

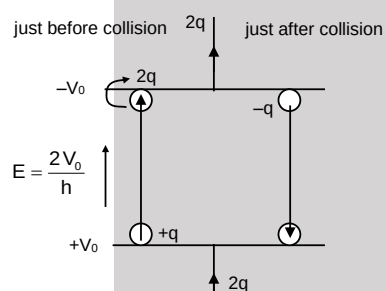
$$I = 2I_g \frac{R_c}{R}$$

$$I_2 = I_g \left(1 + \frac{4R_c}{R} \right)$$

We can see $I_1 > I_2$



15.



$$\frac{Kq}{r} = V_0 = q = \frac{V_0 r}{K}$$

$$\frac{1}{2} \left(\frac{qE}{m} \right) t^2 = h \quad ; \quad \frac{1}{2} \frac{V_0 r}{K} \frac{2V_0}{hm} t^2 = h \quad ; \quad t^2 = \frac{h^2}{V_0^2} \frac{mk}{r}$$

$$t = \frac{h}{V_0} \sqrt{\frac{mk}{r}}$$

During every collision $2q$ charge will flow from circuit.

$$\text{Average current } I_{\text{avg}} = \frac{2q}{t} = \frac{2V_0^2}{h} \frac{r\sqrt{r}}{mk\sqrt{k}}$$

$$I_{\text{avg}} \propto V_0^2$$

14. The balls will bounce back to the bottom plate carrying the opposite charge they went up with.

15. $I_{\text{avg}} \propto V_0^2$

16. $N = 50$

$$A = 2 \times 10^{-4} \text{ m}^2, C = 10^{-4}, R = 50 \Omega, B = 0.02 \text{ T}, \theta = 0.2 \text{ rad}$$

$$N \Phi_{AB} = C \theta$$

$$i_g = \frac{C \theta}{N A B} = \frac{10^{-4} \times 0.2}{50 \times 2 \times 10^{-4} \times 0.02} = 0.1 \text{ A}$$

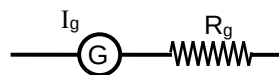
$$i_g \times G = (i - i_g) S$$

$$0.1 \times 50 = (1 - 0.1) \times S$$

$$5 = 0.9 \times S \quad ; \quad S = \frac{50}{9} \Omega$$



17.

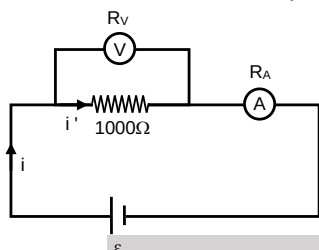


$$V = 100 \times 10^{-3} \text{ V}$$

$$V = I_g(R_g + R_v)$$

$$\frac{10^{-1}}{2 \times 10^{-6}} = R_g + R_v$$

$$5 \times 10^4 \Omega \approx R_v \quad (R_v < 10^5 \Omega)$$



$$I_g R_g = (I - I_g) S$$

$$S = \frac{2 \times 10^{-6} \times 10}{10^{-3} - 2 \times 10^{-6}}$$

$$S = 2 \times 10^{-5} \times 10^3$$

$$\Rightarrow 2 \times 10^{-2}$$

$$\Rightarrow 20 \text{ m}\Omega$$

$$R_A = \frac{20 \times 10^{-3} \times 10}{10} = 20 \times 10^{-3} \Omega$$

$$i = \frac{\varepsilon}{\left(\frac{1000 \times 50 \times 10^3}{51 \times 10^3} \right)} = \frac{51\varepsilon}{5 \times 10^4} \quad (\because R_A \rightarrow 0)$$

$$i' = i \left(\frac{R_v}{51 \times 10^3} \right) = \frac{\varepsilon}{1000}$$

$$\text{measured resistance } \therefore R_m = \frac{i' \times 1000}{i} = \frac{\varepsilon}{51\varepsilon} \times 5 \times 10^4 = \frac{5 \times 10^4}{51} = 980.4 \Omega$$

PART - II

1. Let R be their individual resistance at 0°C . Their resistance at any other temperature t is

$$R_1 = R(1 + \alpha_1 t) \text{ and } R_2 = R(1 + \alpha_2 t).$$

In series

$$R_{\text{series}} = R_1 + R_2 = R[2 + (\alpha_1 + \alpha_2)t]$$

$$= 2R \left[1 + \frac{\alpha_1 + \alpha_2}{2} t \right] \quad \alpha_{\text{series}} = \frac{\alpha_1 + \alpha_2}{2}$$

$$\text{In Parallel} \quad R_{\text{parallel}} = \frac{R_1 R_2}{R_1 + R_2} = \frac{R(1 + \alpha_1 t) R(1 + \alpha_2 t)}{R(2 + \alpha_1 + \alpha_2)t}$$

$$\approx \frac{R^2(1 + (\alpha_1 + \alpha_2)t)}{2R(1 + \frac{\alpha_1 + \alpha_2}{2}t)} \approx \left(1 + \frac{\alpha_1 + \alpha_2}{2}t \right)$$

$$\alpha_{\text{parallel}} = \frac{\alpha_1 + \alpha_2}{2}.$$





2. $R = \frac{\rho \ell}{A}$ ($\because V = A\ell$ const)

$$V = A\ell$$

By differentiation $0 = \ell dA + Ad\ell$ (1)

By differentiation $dR = \frac{\rho(Ad\ell - \ell dA)}{A^2}$ (2)

$$dR = \rho \frac{2Ad\ell}{A^2}, \quad dR = \frac{2\rho d\ell}{A}$$

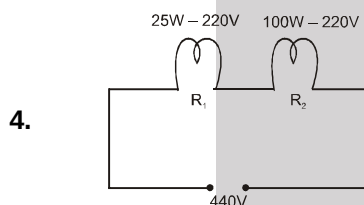
$$\text{or } \frac{dR}{R} = 2 \cdot \frac{d\ell}{\ell}$$

$$\text{So, } \frac{dR}{R} \% = 2 \cdot \frac{d\ell}{\ell} \% = 2 \times 0.1\%$$

$$\frac{dR}{R} \% = 0.2\% \quad \text{Ans.}$$

3. $x = \frac{V}{\ell} = \frac{IR}{\ell} = \frac{IR}{\ell} \left(\frac{\rho \ell}{A} \right) = \frac{I\rho}{A}$

$$x = \frac{0.2 \times 4 \times 10^{-7}}{8 \times 10^{-7}} = \frac{0.8}{8} = 0.1 \text{ V/m.}$$



$$\text{As } R_1 = \frac{220}{25} \times 220 \quad \text{and } R_2 = \frac{220}{100} \times 220$$

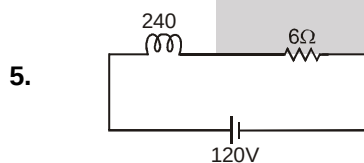
$$R = R_1 + R_2$$

$$= 220 \times 220 \left(\frac{1}{25} + \frac{1}{100} \right)$$

$$= 220 \times 220 \frac{1}{20}$$

$$\therefore I_{\text{live}} = \frac{440}{\frac{220 \times 220}{20}} = \frac{40}{220} \text{ A}$$

$\therefore$ 1st bulb (25 W) will fuse only



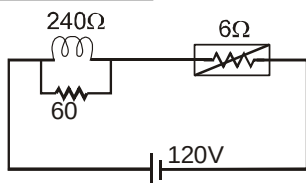
$$P = \frac{V^2}{R}$$

$$R = \frac{120 \times 120}{60} = 240 \Omega$$

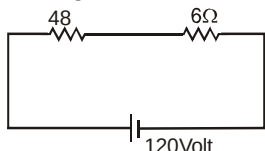
$$R_{\text{eq}} = 240 + 6 = 246 \Omega$$

$$V_1 = \frac{240}{246} \times 120 = 117.073 \text{ volt}$$





$$V_2 = \frac{48}{54} \times 120 = 106.66 \text{ Volt}$$



$$V_1 - V_2 = 10.04 \text{ Volt}$$

6. Statements I is false and Statement II is true

7. Total power (P) = $(15 \times 40) + (5 \times 100) + (5 \times 80) + (1 \times 1000) = 2500W$

$$P = VI$$

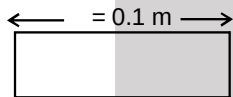
$$\Rightarrow I = \frac{2500}{220} \text{ A}$$

$$= \frac{125}{11}$$

$$= 11.3 \text{ A}$$

Minimum capacity should be 12 A

8.



$$+ \quad V = 5V \quad -$$

$$v_d = 2.5 \times 10^{-4} \text{ m/s}$$

$$n = 8 \times 10^{28} / \text{m}^3$$

$$I = ne A v_d$$

$$\frac{VA}{\rho \ell} = ne A v_d$$

$$\rho = \frac{V}{nev_d \ell} = \frac{5}{8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-4} \times 0.1}$$

$$= 1.6 \times 10^{-5} \Omega \text{ m}$$

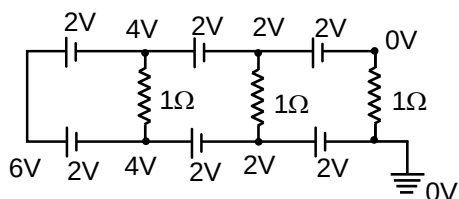
9.

$$S = \frac{i_g G}{I - i_g}$$

$$\text{here } i_g = 10^{-3} \text{ A} \quad G = 10^2 \Omega, \quad I = 10 \text{ A}$$

$$S \approx 10^{-2} \Omega$$

10.



p.d. across each resistance is zero so current is also zero.

11.

In a balanced Wheatstone bridge if the position of cell and galvanometer is exchanged the null point remains same.



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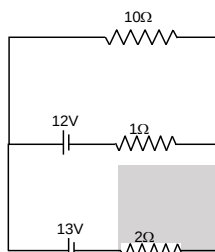
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12. Full deflection current, $I_g = 5\text{mA}$
Resistance of galvanometer, $G = 15\Omega$.

$$\begin{aligned} R &= \frac{V}{I_g} - G \\ &= \frac{10}{5 \times 10^{-3}} - 15 \\ &= 2000 - 15 \\ &= 1985 \Omega \\ &= 1.985 \times 10^3 \Omega \end{aligned}$$

13.



$$E_{eq} = \frac{\frac{12}{1} + \frac{13}{1}}{\frac{1}{1} + \frac{2}{2}}$$

$$= \frac{\frac{37}{2}}{\frac{3}{2}} = \frac{37}{3}$$

$$r_{eq} = \frac{2}{3}$$

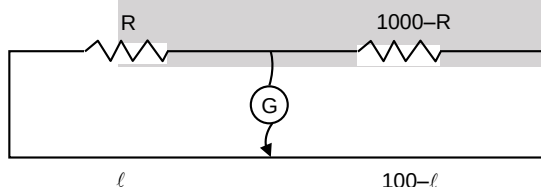
$$I = \frac{\frac{37}{3}}{\frac{2}{3} + 10} = \frac{\frac{37}{3}}{\frac{32}{3}} = \frac{37}{32}$$

$$\text{Voltage across load} = IR = \left(\frac{37}{32}\right)(10) = 11.56\text{V}$$

14.

$$\begin{aligned} \text{Using formula } r &= R \left(\frac{\ell_1}{\ell_2} - 1 \right) \\ &= 5 \left(\frac{52}{40} - 1 \right) = 5 \times \frac{12}{40} = 1.5\Omega \end{aligned}$$

15.



Say resistances are R and $1000 - R$

$$\text{For case-I} \quad \frac{R}{\ell} = \frac{1000 - R}{100 - \ell}$$

$$\text{For case-II} \quad \frac{1000 - R}{\ell - 10} = \frac{R}{110 - \ell}$$

Multiplying both equation





$$\frac{R(1000 - R)}{\ell(\ell - 10)} = \frac{(1000 - R)R}{(100 - \ell)(110 - \ell)}$$

$$\Rightarrow \ell^2 - 10\ell = 11000 + \ell^2 - 210\ell$$

$$\Rightarrow 200\ell = 11000$$

$$\Rightarrow \ell = 55 \text{ cm}$$

Putting in first equation

$$\frac{R}{55} = \frac{1000 - R}{45}$$

$$45R = 55000 - 55R$$

$$R = 550 \Omega$$

16.

Color Codes	Values	Multiplier	Tolerance (%)
Black	0	1	
Brown	1	10	
Red	2	100	
Orange	3	1K	
Yellow	4	10K	
Green	5	100K	
Blue	6	1M	
Violet	7	10M	
Grey	8	100M	
White	9	1G	
Gold			5
Silver			10

17.

In circuit, when $R_h = 2\Omega$

$$i_1 = \frac{6}{4 + 2} = 1A, \varepsilon = \frac{1 \times 4}{AB} \times AJ$$

$$AJ = \frac{AB}{4} \times \varepsilon \quad \dots(1)$$

when $R_h = 6\Omega$

$$i_2 = \frac{6}{4 + 6} = 0.6A$$

$$\varepsilon_2 = \frac{0.6 \times 4}{AB} \times AJ$$

$$AJ = \frac{\varepsilon_2 \times AB}{0.6 \times 4} \quad \dots(2)$$

From (1) & (2)

$$\frac{\varepsilon_2 \times AB}{0.6 \times 4} = \frac{\varepsilon \times AB}{4}$$

$$\varepsilon_2 = 0.6 \times \varepsilon$$

$$= 0.6 \times 0.5 = 0.3V$$



18. $dR = \frac{cd\ell}{\sqrt{\ell}}$

According to Question

$$\int_0^{\ell} c \frac{d\ell}{\sqrt{\ell}} = \int_{\ell}^1 c \frac{d\ell}{\sqrt{\ell}}$$

$$(2\sqrt{\ell})_0^{\ell} = (2\sqrt{\ell})_{\ell}^1$$

$$2\sqrt{\ell} = 2 - 2\sqrt{\ell}$$

$$4\sqrt{\ell} = 2$$

$$\ell = \frac{1}{4} = 0.25 \text{ m.}$$

19. $Q_0 \propto i_G \Rightarrow Q_0 C = i_G$

I-case $CQ_0 = \frac{V}{220 + R} \dots (1)$

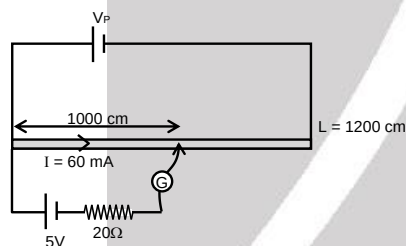
II-case $C \frac{Q_0}{5} = \frac{V}{\left(220 + \frac{5R}{5+R}\right)} \times \frac{5}{5+R} \dots (2)$

from (1) and (2) $R = 22\Omega$

20. Total power is $(15 \times 45) + (15 \times 100) + (15 \times 10) + (2 \times 1000)$
 $= 4325 \text{ W}$

So current is $= \frac{4325}{220} = 19.66 \text{ A}$

Ans is 20 Amp.



21.

Potential gradient $= \frac{5}{1000} = \frac{V_P}{1200}$
 $V_P = 6V$

and $R_P = \frac{V_P}{I} = \frac{6}{60 \times 10^{-3}} = 100\Omega$

22. $V_g = i_g R_g = 0.1 \text{ V}$
 $V = 10 \text{ V}$

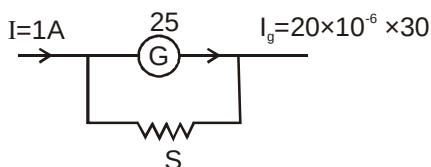
$$R = R_g \left(\frac{V}{V_g} - 1 \right)$$

$$= 100 \times 99 = 9.9 \text{ K}\Omega$$



HIGH LEVEL PROBLEMS (HLP)

1.



$$I_g = 20 \times 10^{-6} \times 30 = 0.6 \times 10^{-3} \text{ A}$$

As we know

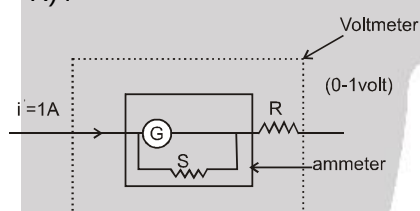
$$I_g R_g = S(I - I_g)$$

$$25 \times 0.6 \times 10^{-3} = S \times (1 - 0.6 \times 10^{-3})$$

$$S = \frac{15.0 \times 10^{-3}}{1 - 0.6 \times 10^{-3}} = 0.015 \Omega \quad \text{Ans.}$$

For voltmeter

$$V = (R_A + R) i$$



Resistance of ammeter is

$$R_A = \frac{S \times R_g}{S + R_g}$$

$$R_A \sim S = 0.015$$

$$V = (R_A + R) i$$

$$1 = (0.015 + R) \times 1$$

$$R = 0.985 \Omega$$

Ans.

2.

$$4 = i \times 10 \times 10^3.$$

$$i = \frac{4}{10^4} = \frac{120}{X + 10^4}$$

$$X + 10^4 = 30 \times 10^4$$

$$X = 29 \times 10^4 \quad \text{Ans.}$$

3.

(a)

$$P = 40 \text{ W}$$

$$h = 10 \text{ m}$$

$$V = 200 \text{ litre}$$

$$t = ?$$

$$\eta = 90 \%$$

$$\frac{mgh}{t} \times \eta = P$$

$$\frac{m}{t} = \frac{40}{10 \times 10 \times 0.9} = \frac{4}{9} \text{ kg/s.}$$

Ans.

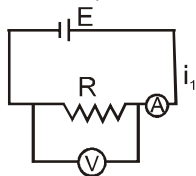
$$(b) \quad m = 200 \times 10^3 \times 10^{-3} \text{ kg}$$

$$= \frac{4}{9} t$$

$$t = \frac{1800}{4} = \frac{900}{2} = 450 \text{ sec.} \quad \text{Ans.}$$



4. $E = 3.4 \text{ volt}$, $r = 3 \Omega$, $R_A = 2 \Omega$, $R = 100 \Omega$.



$$i_1 = 0.04 \text{ A}, \quad V = ?, \quad R_V = ?$$

$$0.04 = \frac{3.4}{3 + 2 + \frac{100 R_V}{100 + R_V}}$$

$$5 + \frac{100 R_V}{100 + R_V} = \frac{3.4}{0.04} = 85$$

$$R_V = 400 \Omega \quad \text{Ans.}$$

$$V = 0.04 \times \frac{100 \times 400}{500}$$

$$V = 3.20 \text{ V} \quad \text{Ans.}$$

For ideal voltmeter

$$R_V \rightarrow \infty$$

$$i = \frac{3.4}{3 + 2 + 100} = \frac{3.4}{105}$$

$$V = i \times 100 = \frac{3.4}{105} \times 100 = \frac{68}{21} \text{ V} \quad \text{Ans.}$$

$$\begin{aligned} 5. \quad I_1 &= \frac{E}{r + R} \Rightarrow r + R = \frac{E}{I_1} \\ I_2 &= \frac{2E}{2r + R} \Rightarrow 2r + R = \frac{E}{I_2} \\ I_3 &= \frac{E}{\frac{r}{2} + R} \Rightarrow \frac{r}{2} + R = \frac{E}{I_3} \end{aligned}$$

To show that $3 I_2 I_3 = 2 I_1 (I_2 + I_3)$

$$\text{L.H.S} = 3 \times \frac{2E}{2r + R} \times \frac{E}{\frac{r}{2} + R} = 2 \frac{E}{r + R} \left(\frac{2E}{2r + R} + \frac{E}{\frac{r}{2} + R} \right) = \text{R.H.S}$$

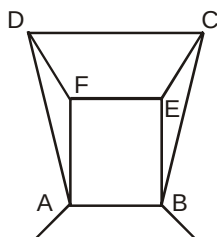
$$3E = \frac{1 \times E}{r + R} \left\{ 2 \left(\frac{r}{2} + R \right) + 2r + R \right\} \Rightarrow 3r + 3R = r + 2R + 2r + R$$

Hence its prove.

$$\begin{aligned} 6. \quad \frac{nr}{m} &= R \Rightarrow I = \frac{nE}{2R} \\ I' &= \frac{mE}{\frac{mr}{n} + R} = \frac{mE}{\frac{m}{n} \times \frac{m}{n} R + R} = \frac{mE \times n^2}{R(m^2 + n^2)} = \frac{mn^2}{(m^2 + n^2)} \times \frac{2I}{n} \end{aligned}$$

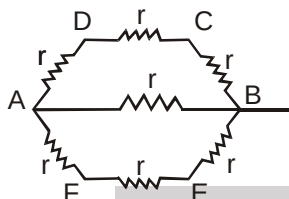


7.



By symmetry

Current in branches FD and CE are zero. Because potential difference across them is zero



$$R_{eq} = \frac{3r}{5}$$

Ans.

8.

$$R_g = 100 \Omega$$

$$I_g = 1 \text{ mA}, \quad S = ?, \quad I = 10 \text{ A}, \quad P = 1 \text{ W}$$

$$100 \times 10^{-3} = (10 - 10^{-3}) S$$

$$S = \frac{0.1}{10 - 10^{-3}} \approx 0.01 \quad \text{Ans.}$$

$$P = (10 - 10^{-3})^2 \times S = (10 - 10^{-3}) \times 0.1 \\ \approx 10 \times 0.1 = 1 \text{ Watt}$$

 $\therefore$ Power dissipated is slightly less than 1W so its use is safe

Ans.

$$4 = \frac{E}{\frac{100 \times S}{100 + S} + r} \quad \dots\dots\dots(i), \quad 4 = \frac{E}{\frac{100 \times 0.01}{100} + r}$$

$$1 = \frac{E}{\left(r + \frac{100 \times S}{100 + S} + 1.5\right)} \quad \dots\dots\dots(ii), \quad \frac{1}{100} + r = \frac{2}{4}$$

From (i) & (ii)

$$1 = \frac{E}{\frac{E}{4} + 1.5}, \quad r = 0.49 \Omega \quad \text{Ans.}$$

$$E = 2 \text{ V}$$

Ans.

9.

$$(a) \quad V_A = 6 \text{ V}, \quad V_C = 2 \text{ V}$$

Ans.

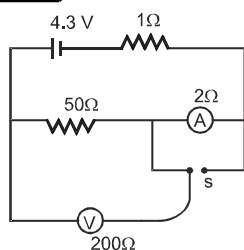
$$(b) \quad E = x \ell \Rightarrow 4 = \ell \times \frac{6}{100} \Rightarrow \ell = \frac{200}{3} \text{ cm} \quad \text{Ans.}$$

$$(c) \quad \text{In secondary circuit current is zero} \quad \text{Ans.}$$

$$(d) \quad 6 \text{ V}, 6 - 7.5 = -1.5 \text{ V}, \text{ no such point D exists} \quad \text{Ans.}$$



10. (a)



$$R_{eq} = 1 + 2 + \frac{50 \times 200}{250}$$

$$= 43$$

$$i = \frac{4.3}{43} = 0.1 \text{ A}$$

Ans.

$$V = 40 \times 0.1 = 4 \text{ V}$$

Ans.

$$(b) R_{eq} = 1 + \frac{52 \times 200}{252}$$

$$= \frac{252 + 10400}{252}$$

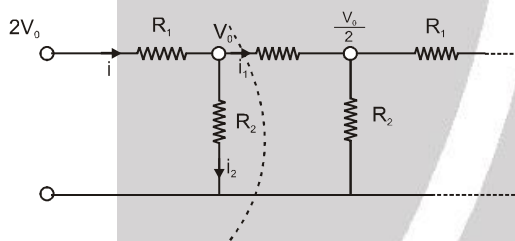
$$i = \frac{4.3 \times 252}{252 + 10400} = \frac{1083.6}{10652} \Rightarrow i_A = \frac{1083.6}{10652} \times \frac{200}{200 + 52} = 0.08 \text{ A}$$

Ans.

$$V = 4.3 - i \times 1 = 4.2 \text{ V}$$

Ans.

11.



$$i = i_1 + i_2$$

$$\Rightarrow \frac{V_0}{R_2} + \frac{V_0}{2R_1} = \frac{V_0}{R_1} \Rightarrow \frac{1}{R_2} = \frac{1}{R_1} - \frac{1}{2R_1} \Rightarrow \frac{R_1}{R_2} = \frac{1}{2}$$

$$R_2 = 2R_1$$

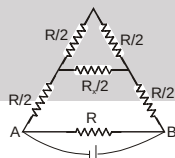
$$R_{AB} = R_{eq} = R_1 + \frac{2R_1 R_{eq}}{2R_1 + R_{eq}}$$

On solving

$$R_{eq}^2 - R_1 R_{eq} - 2R_1^2 = 0 ; \quad R_{eq} = 2R_1$$

12.

It follows from symmetry considerations that the initial circuit can be replaced by an equivalent one (as shown).



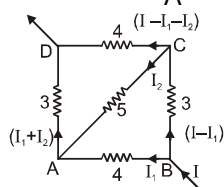
We replace the inner triangle consisting of an infinite number of elements by a resistor of resistance $R_{AB}/2$, where the resistance R_{AB} is such that $R_{AB} = R_x$ and $R_{AB} = ap$. After simplification, the circuit becomes a system of series and parallel connected conductors. In order to find R_x , we write the equation

$$R_x = R \left(R + \frac{RR_x/2}{R + R_x/2} \right) \left(R + R + \frac{RR_x/2}{R + R_x/2} \right)^{-1}$$

$$\text{Solving the equation, we obtain } R_{AB} = R_x = \frac{R(\sqrt{7}-1)}{3} = \frac{ap(\sqrt{7}-1)}{3}$$



13. As, $R = \frac{\rho \ell}{A}$



i.e., resistance of wire is proportional to its lengths other factors being constant. So, resistance of wire.
 AB or $CD = 4$ unit

Resistances of arm BC or $AD = 3$ unit

and resistance of diagonal $AC = 5$ unit

(Taking proportionality constant to be unity) Resistance R_1 between AC is parallel combination of R_{ADC} , R_{AC} and R_{ABC} .

So, $\frac{1}{R_1} = \frac{1}{7} + \frac{1}{5} + \frac{1}{7}$ i.e., $R_1 = \frac{35}{17}$ unit

Applying Kirchoff's laws for closed meshes $BCAB$ and $ADCA$,

$$3(I - I_1) + 5I_2 - 4I_1 = 0$$

$$3I - 7I_1 + 5I_2 = 0 \quad \text{.....(i)}$$

and $3(I_1 + I_2) - 4(I - I_1 - I_2) + 5I_2 = 0$
 $-4I + 7I_1 + 12I_2 = 0 \quad \text{.....(ii)}$

Adding Eqs.(i) and (ii),

$$I_2 = \frac{I}{17}$$

Substituting for I_2 in Eq.(i)

$$I_1 = \frac{8I}{17}$$

If R_2 is the effective resistance between points B and D ,

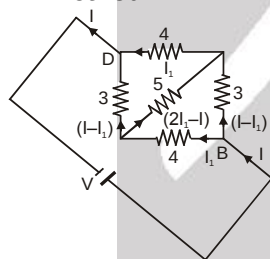
$$IR_{BD} = 4I_1 + 3(I_1 + I_2) = 7I_1 + 3I_2$$

Substituting for I_1 and I_2 ,

$$IR_{BD} = 7 \cdot \frac{8I}{17} + \frac{3I}{17}, \quad \text{i.e.} \quad R_{BD} = \frac{59}{17} \text{ units}$$

$$\frac{R_{BD}}{R_{AC}} = \frac{59}{35} \quad \text{Ans.}$$

2nd method



$$-4I_1 - 5(2I_1 - I) + 3(I - I_1) = 0 \quad \text{.....(i)}$$

$$-V + 3(I - I_1) + 4I_1 = 0 \quad \text{.....(ii)}$$

(i) and (ii) $\Rightarrow V = \frac{59}{17} I$

$$\Rightarrow R_{BD} = \frac{59}{17} \text{ units}$$



14. (a) Let thermal energy Q_A generated when coil A is switched on

$$Q_A = \frac{V^2}{R_A} t_A = \frac{V^2}{R_A} 10$$

Let thermal energy Q_B generated when coil B is switched on

$$Q_B = \frac{V^2}{R_B} t_B = \frac{V^2}{R_B} 20$$

since $Q_A = Q_B$

$$\text{we have } \frac{V^2}{R_A} 10 = \frac{V^2}{R_B} 20$$

$$\text{or } 2R_A = R_B$$

$$\text{when A \& B are connected in series thermal energy } Q_S = \frac{V^2}{R_{eq}} t_S = \frac{V^2}{(R_A + R_B)} t$$

$$Q_S = Q_P = Q_A = Q_B$$

$$\therefore \frac{V^2 t_S}{R_A + R_B} = \frac{V^2}{R_A} 10$$

$$\text{or } t_S = 10 \frac{(R_A + R_B)}{R_A} = \frac{3R_A}{R_A} 10 = 30 \text{ minutes}$$

- (b) when A & B are in parallel

$$Q_P = \frac{V^2}{R'_{eq}} t_P = \frac{V^2 t_P (R_A + R_B)}{(R_A R_B)}$$

$$Q_P = Q_A$$

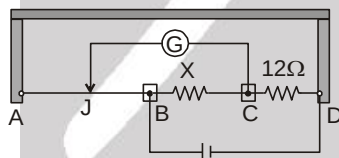
$$\text{or } \frac{V^2 t_P (R_A + R_B)}{(R_A R_B)} = \frac{V^2}{R_A} 10$$

$$\text{or } t_P \frac{3R_A}{2R_A^2} = \frac{10}{R_A}$$

$$\text{or } t_P = \frac{20}{3} \text{ minutes}$$

15. (a) There are no positive and negative terminals on the galvanometer because only zero deflection is needed.

- (b)



- (c)

$$AJ = 60 \text{ cm}$$

$$\therefore BJ = 40 \text{ cm}$$

If no deflection is taking place. Then the Wheatstone bridge is said to be balanced. Hence

$$\frac{X}{12} = \frac{R_{BJ}}{R_{AJ}}$$

$$\text{or } \frac{X}{12} = \frac{40}{60} = \frac{2}{3}$$

$$\text{or } X = 8\Omega$$

Ans.



$$I_g = \frac{E}{R+r+\frac{GS}{G+S}} \cdot \frac{S}{G+S} = \frac{ES}{(R+r)(G+S)+GS}$$

for $S = 10\Omega$, $I_g = \frac{9}{50} \times 200 \times 10^{-3} = 36 \times 10^{-3} \text{ amp}$

$$36 \times 10^{-3} = \frac{10E}{100(10+G)+10G} \quad \dots(1)$$

for $S = 50\Omega$, $I_g = \frac{30}{50} \times 200 \times 10^{-3} = 120 \times 10^{-3} \text{ amp}$

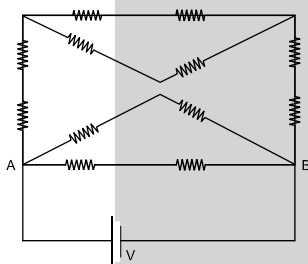
$$120 \times 10^{-3} = \frac{50E}{100(50+G)+50G} \quad \dots(2)$$

from (1) and (2)

$$\frac{36}{24} = \frac{100(50+G)+50G}{100(10+G)+10G} \Rightarrow G = \frac{700}{3} \Omega \quad \text{Ans.}$$

from eq (1) $E = 96 \text{ V}$ **Ans.**

19.

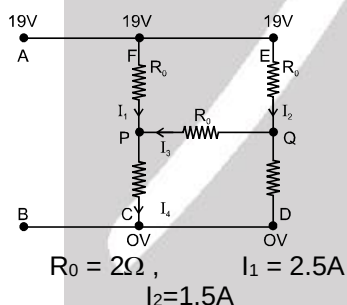


$$R_{AB} = \frac{5}{6} R$$

$$\text{Power} = \frac{V^2}{R_{AB}} = \frac{V^2}{\frac{5}{6}R} = \frac{6}{5}P = \frac{3xP}{5} \Rightarrow x = 2$$

20.

(A)



$$V_P = 19 - R_0 I_1 = 19 - (2) \times (2.5)$$

$$V_P = 14 \text{ V}$$

Similarly

$$V_Q = 16 \text{ V}$$

$$\Rightarrow I_3 = \frac{V_Q - V_P}{R_0} \Rightarrow I_3 = 1 \text{ A}$$

$$(B) I_4 = \frac{V_P - V_C}{R} = I_1 + I_3$$

$$\Rightarrow I_4 = \frac{14 - 0}{R} = I_1 + I_3 \Rightarrow \frac{14}{R} = 2.5 + 1$$

$$R = 4\Omega$$





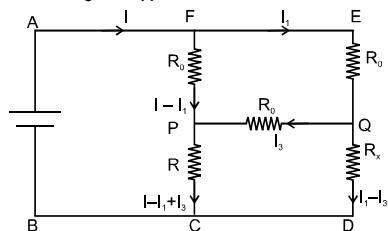
(C) Applying KVL in FEQPF

$$-I_1 R_0 - I_3 R_0 + (I - I_1) \times R_0 = 0$$

$$I = I_3 + 2I_1 \quad \dots (i)$$

Applying KVL in AFEDCBA ; $V - I_1 R_1 - (I_1 - I_3) R_X = 0$

$$I_1 = \frac{V + I_3 R_X}{R_0 + R_X} \quad \dots (ii)$$



Applying KVL in AFCBA ; $V - (I - I_1) R_0 - \frac{V + I_3 R_X}{R_0 + R_X} (I - I_1 + I_3) R = 0$

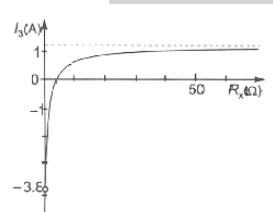
$$V - I(R_0 + R) + I_1(R_0 + R) - I_3 R = 0$$

$$V - (I_3 + 2I_1)(R_0 + R) + I_1(R_0 + R) - I_3 R = 0 \text{ Using (i)}$$

$$V - I_3(R_0 + 2R) - I_1(R_0 + R) = 0$$

$$V - I_3(R_0 + 2R) - \frac{V + I_3 R_X}{R_0 + R_X} (R_0 + R) = 0 \text{ Using (ii)}$$

$$I_3 = \frac{V(R_X - R)}{(2R_0 + 3R)R_X + R_0(R_0 + 2R)}$$



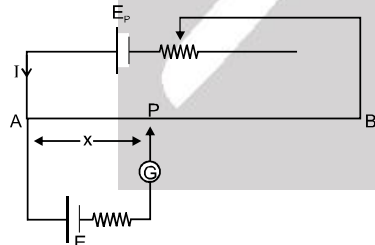
The graph of the function is a hyperbola. Its special points are: at $R_X = 0$ $I_3 = -3.8$ A ; at $R_X = 32 \Omega$ $I_3 = 1$ A ; If $R_X \rightarrow \infty$, I_3 tends to $19/16 = 1.1875$ A.

21.

$$\lambda = 2x$$

$$R_{AB} = \int_0^1 \lambda dx$$

$$R_{AB} = 1 \Omega$$



$$R_{AP} = x^2$$

$$I = \frac{E_P}{R + R_{AB}} = \frac{E_P}{1 + t^2}$$

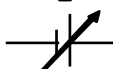
$$\text{Now } V_{AP} = IR_{AP} = E_S$$

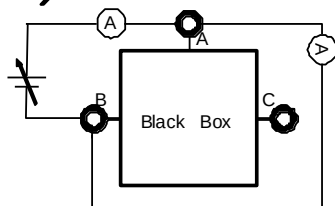
$$\Rightarrow E_S = \frac{E_P x^2}{(1 + t^2)} \Rightarrow \text{at } t = 1 \text{ sec, } x = 1/2 ; E_S = \frac{E_P \left(\frac{1}{2^2} \right)}{1 + 1} = \frac{E_P}{8} \Rightarrow \gamma = 8$$



22. (a)  Ammeter

 Voltmeter

 Variable Power supply



(b) See graphs for the calculations of slopes.

$R_{AB} = 0.98 \text{ k}\Omega$, $R_{BC} = 4.60 \text{ k}\Omega$, $R_{CA} = 5.67 \text{ k}\Omega$

